Scale without Scope: Firm Specialization and Rising Industry Concentration

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Abstract

U.S. industries have become increasingly concentrated, with large firms accounting for a growing share of economic activity. Using U.S. manufacturing firm-product data from 1990 to 2020, I show that rising concentration is accompanied by increasing specialization among industry leaders. The average number of products produced by the top 1% of firms declines by more than 40%. I develop and estimate an oligopolistic model of multi-product firms, in which firms choose the product scope and invest in scalable, product-specific technology investments. As these technologies become more efficient, firms have stronger incentives to invest in their most productive products, while external competition and rising internal investment costs make peripheral products less attractive. Firms can therefore grow larger while producing fewer products. I structurally estimate the model using the U.S. firm micro data. Quantitatively, improvements in scalable technology efficiency generate increases in industry concentration and narrower product portfolios, consistent with the observed trends.

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1 Introduction

U.S. industries have become increasingly concentrated over the past four decades, with a growing share of economic activity accounted for by a small number of large firms (Autor et al. 2023). What does this rise in concentration imply for the level of competition within markets? And what are its consequences for aggregate productivity and welfare? Understanding how industry leaders come to account for a larger share of activity is central to assessing both the causes and the consequences of rising concentration.

This paper answers to these questions in three ways. First, I provide new empirical evidence that rising industry concentration is accompanied by changes in firms' product offerings. Leading firms increasingly concentrate production in a smaller number of products. I show that the top 1% of firms produce substantially fewer products in 2020 than in 1990. Second, I develop a model of multi-product firms in which technological progress in scalable fixed-cost investments changes the relationship between firm scale and product scope. This technological progress allows firms to become larger while narrowing their product portfolios. Third, I estimate the framework using U.S. firm-level data. I then quantify the implications of this transformation for aggregate productivity and welfare.

In the first part of the paper, I examine how the product offerings of leading U.S. firms have changed over time. Using detailed firm-product data covering a near-universe of U.S. manufacturing firms from 1990 to 2020, I document a substantial decline in product scope among industry leaders. The employment-weighted average number of products produced by the top 1% of firms falls by more than 40% over this period. Using the panel structure of the data, I find that more than half of this decline reflects continuing industry leaders reducing the number of products they produce. By contrast, firms with narrower product portfolios moving into the top 1% of the firm-size distribution account for a smaller share of the decline. These findings motivate the model's focus on why leading firms narrow their own product portfolios, even as they account for a growing share of industry activity.

In the second part of the paper, I develop a general equilibrium model of multi-product firms to rationalize these facts and study their productivity and welfare implications. Firms choose how many products to develop, the production technology for each product, and their scale of production. Firms are heterogeneous in fundamental productivity, and their product-level productivity exhibits a core-periphery structure: they are more productive in their core products and less efficient in producing varieties outside their core competence.

In the model, firms choose their production technology by making product-specific investments that lower marginal production costs. In the spirit of Sutton (1991), these investments represent endogenous fixed costs incurred upfront to improve production efficiency. Once made, the same investment can support a larger volume of production without a proportional increase in expenditure. Technology investment is therefore scalable within each product line.[1]

Firms make decisions in two stages. In the first stage, they choose their product scope, which determines how many product markets to enter. In the second stage, they choose product-level technology investment and compete à la Bertrand within each product market. Conditional on product scope, technology investment determines a firm's marginal cost and therefore its scale of production in each market. The resulting profits determine the value of operating a given product portfolio and feed back into firms' scope choices in the first stage. In equilibrium, these firm-level decisions jointly determine product-market participation, markups, and aggregate productivity.

Two forces govern firms' technology investment in each product. On the benefit side, investment reduces unit labor requirements, generating greater savings over a larger production base. Producers with higher fundamental productivity therefore have stronger incentives to invest, as they tend to operate at a larger scale. On the cost side, a common convex firm-level investment cost makes products compete for investment resources. Expanding investment in one product raises the marginal cost of upgrading the rest of the portfolio. Thus, investment in each product depends not only on its competitive position, but also on investment opportunities elsewhere in the firm's portfolio.

When the gains from technology investment are small, operating a broad product portfolio and producing more varieties is an important reason why large firms are large, as in Klette and Kortum (2004) and Bernard et al. (2010). As the technology becomes more effective, the return to investing in a firm's stronger products rises, allowing the firm to strengthen its position in those markets. At the same time, investment in peripheral products becomes less attractive as stronger competitors invest more in those markets. The within-firm linkage further reinforces this response: as the firm's total investment rises, so does its marginal cost of investment, further reducing the value of investing in peripheral products. As a result, leading firms can become more productive and account for larger shares in the product markets they continue to serve, while operating in fewer product markets.

Lastly, I estimate a quantitative version of the model and assess the contribution of

[1] Examples include programmable automation, digital process engineering, and specialized machine-tool programming.

rising efficiency of scalable technology to aggregate concentration, productivity, and welfare. I estimate the key parameters using Simulated Method of Moments (SMM). The estimation targets the firm-employment distribution and the level and size gradient of the revenue-to-variable-cost ratio. I then discipline the historical change in technology efficiency by matching the proportional increase in the aggregate fixed-cost share between 1990 and 2020.

Quantitatively, I find that improvements in IT efficiency between 1990 and 2020 generate increases in industry concentration, markups, and productivity dispersion consistent with observed patterns in the U.S. economy. These improvements also reduce the product scope of leading firms. Despite the increase in concentration, aggregate productivity rises by 8.3% and welfare by 5.7%. These results show that productivity-enhancing technological change can lead to greater concentration and firm specialization while generating positive welfare gains. This finding is consistent with the literature on "good concentration," which emphasizes that rising concentration need not imply welfare losses when driven by productivity improvements (Covarrubias et al. 2020).

I further show that firms' endogenous technology investment and product scope choices contribute to these aggregate gains. Relative to a counterfactual with fixed product scope and uniform productivity growth across products within each firm, allowing firms to adjust their product scope and allocate investment across products raises aggregate productivity gains by 3.20 percentage points and welfare gains by 1.09 percentage points. These results highlight the importance of modeling multi-product firms and their endogenous technology choices when quantifying the aggregate effects of technological progress.

This paper is related to several strands of literature. First, I contribute to the literature on the causes and consequences of rising industry concentration and markups. Existing work has emphasized both demand-side forces (Döpper et al. 2025; Ekerdt and Wu 2024; Gupta 2020) and supply-side forces associated with technological change, including the growing importance of scalable and shared inputs (Argente et al. 2025; Hsieh and Rossi-Hansberg 2023; Oberfield et al. 2024). I contribute to this supply-side literature by studying firm scale and product scope jointly. In my model, firms choose both how much to produce within each product and how many products to offer. This introduces an extensive margin absent from oligopoly models with a fixed set of suppliers (Edmond et al. 2023). As firms adjust their product scope, they enter and exit individual product markets. Technological change therefore affects not only firms' market shares and markups, but also the number of competitors within each market.

This paper is also related to the literature on multiproduct firms. Bernard et al. (2010) and Klette and Kortum (2004), among others, emphasize the importance of the product

scope margin for understanding firm growth. I build on this literature by allowing firms to choose their product-level production technology as well as their product scope. Endogenous technology investment changes the relationship between firm size and product scope, allowing firms to grow larger while producing fewer products. This provides a mechanism for understanding the increasing concentration of industry activity among leading firms with narrower product portfolios.

This paper also contributes to the literature on endogenous returns to scale and scope. For example, Lashkari et al. (2024) studies how information technology affects firms' returns to scale. I examine these forces in an oligopolistic setting in which firms have market power and choose their product scope endogenously. This framework allows me to study how technological change affects market structure and markups, and to quantify its implications for aggregate productivity and welfare.

The remainder of the paper is organized as follows. Section 2 documents rising specialization among industry leaders. Section 3 presents the model, and Section 4 characterizes investment, scope, and product-market outcomes. Section 5 provides empirical evidence on the mechanism. Section 6 presents the calibration and model validation. Section 7 studies the historical technology change and policy counterfactuals. Section 8 concludes.

2 Data and Motivating Facts

In this section, I document a set of facts on the rise of firm specialization between 1990 and 2020. These facts reveal a clear change in how leading firms account for industry market share in 2020 compared with the early 1990s. This shift in the pattern of firm expansion motivates the theoretical framework developed below.

2.1 Data

Dun & Bradstreet Data. I use commercial firm- and establishment-level data from Dun & Bradstreet covering the period 1990–2020. The data cover a near-universe of U.S. firms across a broad range of industries. Throughout the paper, I focus on firms in manufacturing industries. The data provide information on firm structure, the products produced by each establishment, establishment-level employment, industry classifications, and other firm and establishment characteristics. I aggregate the establishment-level information to construct each firm's product portfolio and firm-by-product employment. Appendix

A.1 provides further details on the construction of these variables. I next describe how I define industries, firms, and products.

Industry. I define an industry at the two-digit SIC level, a level of aggregation broadly comparable in granularity to four-digit NAICS industries and commonly used in the literature on industry concentration (De Loecker et al. 2020).

Firm. I define a firm as the set of establishments within a two-digit SIC industry that share the same headquarters DUNS number. Under this definition, different industry divisions of a multi-industry conglomerate are treated as separate firms. For example, GE's activities in electronic equipment manufacturing and transportation equipment manufacturing are counted separately. Following Crane and Decker (2020) and Rossi-Hansberg et al. (2021), I exclude firms with fewer than 10 employees. Appendix B.3 shows that the main results are robust to a broader definition that groups all establishments sharing the same headquarters DUNS number into a single firm, regardless of industry.

Product. A product is defined as an eight-digit SIC code reported in the Dun & Bradstreet data. For example, within electronic equipment manufacturing (SIC 36), "computer logic modules," "solid-state memory," "magnetic bubble memory," and "light-emitting diodes" are examples of products. Table A.1 provides a concrete firm-level illustration by listing the products reported for Intel in the Dun & Bradstreet data from 1990 to 2020 in the electric manufacturing industry.

Our product definition is more disaggregated than the four- or five-digit SIC classifications commonly used in the multiproduct-firm literature (e.g., Bernard et al. 2010; Goldberg et al. 2010). In manufacturing, a four-digit SIC definition yields 455 products, while a five-digit SIC definition yields 1,440. By comparison, our eight-digit definition identifies 9,531 unique manufacturing products over 1990–2020.[2] At the same time, our product definition remains coarser than the narrowly defined product markets often used in the industrial organization literature (e.g., Benkard et al. 2026).

Nielsen Retail Scanner Data. I complement the main analysis with the Kilts NielsenIQ Retail Scanner Data, covering 2006–2015. These data are widely used in industrial organization to study market structure and concentration (e.g., Döpper et al. 2025) and

[2]The same product definition is used by Rossi-Hansberg et al. (2021). The exact set of products varies somewhat from year to year, potentially reflecting the introduction of new products and the obsolescence of existing ones. The total number of products, however, remains stable over time.

in macroeconomics and international trade to study firm and product dynamics (e.g., Hottman et al. 2016). The data report weekly unit sales and revenue at the barcode-by-store level for a large sample of retail chains.

The Nielsen data complement the Dun & Bradstreet data by providing alternative, more disaggregated product definitions. This allows me to construct firm product portfolios using finer classifications, such as Nielsen product groups or product modules, and to examine whether the patterns in product scope continue to hold at a more disaggregated level. Appendix A.2 provides further details on the product definitions used in the Nielsen Scanner data.

2.2 Rising Specialization among Top Firms

2.2.1 Fact 1: Declining product scope among top firms

I first examine how the product scope of leading firms changes between 1990 and 2020. In each industry and year, I define leading firms as the top 1% of firms by employment size. I then calculate the average number of products they produce.[3]

Figure 1 documents a substantial decline in the product scope of industry leaders for the U.S. manufacturing industries. Panel (a) shows that the unweighted average falls from 6.02 products in 1990 to 3.85 in 2020. By comparison, firms outside the top 1% produce roughly one product on average throughout the sample, consistent with most firms being single-product producers (Bernard et al. 2010). The decline is even more pronounced when weighting firms by employment. Panel (b) shows that the employment-weighted average falls from 9.06 to 5.21 products. The corresponding measure for the bottom 99% remains stable. Overall, product scope declines by 36.1% and 42.5% under the unweighted and employment-weighted measures, respectively. Thus, compared with industry leaders in the early 1990s, today's leading firms produce a substantially narrower range of products.[4]

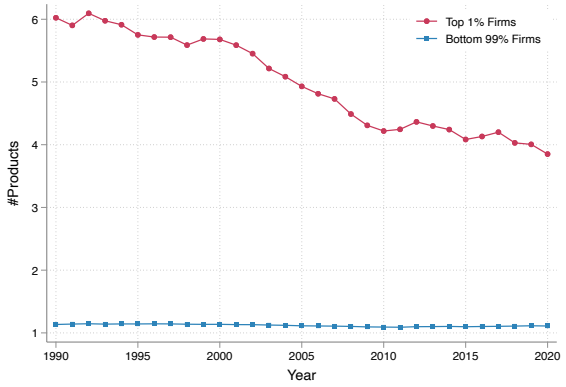
Appendix B.1 shows that the decline in leading firms' product scope also holds when I define industry leaders as the top 20 firms by employment in each industry, a measure commonly used in the literature (e.g., Autor et al., 2020). Under this definition, average product scope declines by 34.6% without employment weights and by 39.6% when

[3]To aggregate these industry-level averages across manufacturing industries, I use each industry's share of total manufacturing employment in 1990 as a fixed weight.

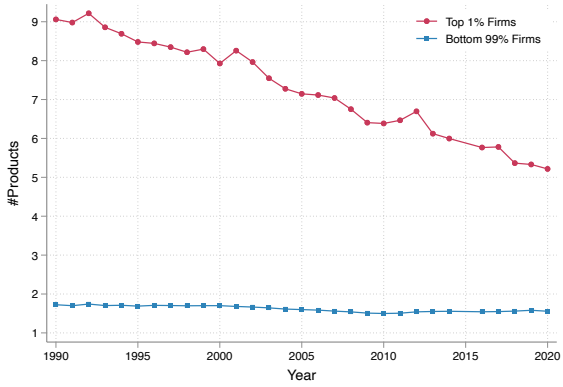
[4]This finding complements Ekerdt and Wu (2024) and Ma (2026) who document declines in the number of six-digit NAICS industries in which manufacturing firms operate. Because this paper focuses on industry concentration, I examine changes in firms' product scope within industries.

Figure 1: Product Scope of Top 1% Firms

(a) Top 1% Firms: Unweighted



(b) Top 1% Firms: Emp weighted



Notes: This figure plots the average number of products produced by the top 1% of firms from 1990 to 2020. A firm is defined as a firm ID by 2-digit SIC industry in the Dun& Bradstreet data. A product is a 8-digit SIC code. For each year, I first compute average product scope among the top 1% of firms within each two-digit SIC industry. I then aggregate these industry-level averages across manufacturing industries, using each industry's share of total manufacturing employment in 1990 as a fixed weight in both panels. Panel (a) plots the resulting aggregate series when firms are equally weighted within each industry, while Panel (b) plots the corresponding series when firms are weighted by their employment. We see that there is a substantial decline in the product scope of top 1% firms.

weighted by employment. Appendix B.2 reports the corresponding results using industry employment shares that vary by year. Where we find average product scope declines by 33.5% without employment weights and by 39.8% when weighted by employment. The results are also robust to an alternative firm definition that counts a firm's products across all manufacturing industries.

2.2.2 Fact II: Declining scope is driven by leading firms shedding products

I next use the panel structure of the data to examine the sources of the decline in product scope among industry leaders between 1990 and 2020. To this end, I decompose changes in employment-weighted average product scope into three components. I find that the decline is driven primarily by continuing top 1% firms reducing the number of products they produce. Employment reallocation toward continuing leaders with narrower product portfolios and changes in the composition of the top 1% through entry and exit also contribute, but to a lesser extent.

Formally, using a decomposition similar to those in Melitz and Polanec (2015) and Ekerdt and Wu (2024), I decompose the decline in employment-weighted average product scope among the top 1% of firms into three components: (i) continuing leaders reducing

the number of products they produce, (ii) employment shifting toward continuing leaders with narrower product portfolios, and (iii) changes in average product scope as firms with different product portfolios enter and exit the top 1%. The first component captures specialization within continuing leaders. The latter two capture changes in the relative importance and composition of firms with different product portfolios, which can reduce average product scope even without individual firms shedding products.

Let N_t^k denote the employment-weighted mean product scope of leading firms in industry k at date t. In the baseline, I define leading firms as the top 1% of firms by employment within each industry. Then,

$$N_t^k = \sum_{i \in \mathcal{L}_t^k} \omega_{it}^k n_{it}^k, \tag{1}$$

where $\mathcal{L}_t^k$ denotes the set of leading firms, n_{it}^k is the number of products produced by firm i, and ω_{it}^k is firm i's employment share among leading firms in industry k: $\omega_{it}^k = \frac{\ell_{it}^k}{\sum_{h \in \mathcal{L}_t^k} \ell_{ht}^k}$. For simplicity, I suppress the industry superscript k throughout this section when there is no risk of confusion.

For a comparison between years t_1 and t_2, let C denote the set of firms that belong to the top 1% in both years. I refer to these firms as continuing leaders. Let E denote firms that enter the top 1% between t_1 and t_2, and let X denote firms that leave the top 1%. The change in employment-weighted product scope can then be written as

$$\begin{aligned} N_{t_2} - N_{t_1} = & \underbrace{\sum_{i \in C} \frac{\omega_{i,t_2} + \omega_{i,t_1}}{2} \left(n_{i,t_2} - n_{i,t_1} \right)}_{\text{Scope adjustment}} + \underbrace{\sum_{i \in C} \frac{n_{i,t_2} + n_{i,t_1}}{2} \left(\omega_{i,t_2} - \omega_{i,t_1} \right)}_{\text{Employment reallocation}} \\ & + \underbrace{\sum_{i \in E} \omega_{i,t_2} n_{i,t_2} - \sum_{i \in X} \omega_{i,t_1} n_{i,t_1}}_{\text{Net entry/exit}} . \end{aligned} \tag{2}$$

The first term fixes firms' employment shares at their average across the two periods while allowing firm i's product scope to change. It captures changes in continuing leaders' contribution that arise from changes in product scope. The term is negative when continuing leaders reduce the number of products they produce. The second term fixes firms' product scope at its average across the two periods while allowing their employment shares to change. It captures changes due to employment reallocation among continuing leaders. The term is negative when employment shifts toward firms with narrower product portfolios. The final term captures the contribution of firms entering the

top 1% net of that of firms leaving the top 1%.

Table 1 reports the decomposition of the decline in employment-weighted product scope between 1990 and 2020. The first row reports results when industry leaders are defined as the top 1% of firms. Their employment-weighted product scope declines by 3.84 products over this period. Scope adjustment among continuing leaders accounts for 2.15 products, or 55.8% of this decline. Thus, more than half of the decline reflects firms that remain in the top 1% reducing the number of products they produce. Employment reallocation accounts for another 0.58 products, or 15.1%, as employment shifts toward continuing leaders with narrower product portfolios. Net entry and exit account for the remaining 1.12 products, or 29.0%. The decomposition therefore shows that an important part of the aggregate decline reflects continuing industry leaders narrowing their own product portfolios.

Table 1: Decomposing changes in top firm product scope

Sample	Scope adjustment (1)	Emp reallocation (2)	Net entry (3)	Total change (4)
Top 1% firms	−2.15 (55.8%)	−0.58 (15.1%)	−1.12 (29.0%)	−3.84
Top 5% firms	−1.46 (50.1%)	−0.70 (24.1%)	−0.75 (25.8%)	−2.92
Top 10% firms	−1.28 (48.6%)	−0.72 (27.2%)	−0.64 (24.2%)	−2.64

Notes: Annual cumulative decomposition of employment-weighted product scope from 1990 to 2020, restricted to manufacturing (SIC2 20–39). Top firms are selected separately within each SIC2 and year. Industry employment weights are fixed at their 1990 values. Parentheses report each component's percentage of the total change, calculated before rounding. A negative percentage indicates an offset to the decline.

The second and third rows repeat the decomposition, defining industry leaders as the top 5% and 10% of firms by employment, respectively. The results are consistent with the top 1% baseline. In both groups, continuing leaders reducing the number of products they produce account for a substantial share of the decline in average product scope.

To recap, I have shown that the product scope of the top 1% of firms declines substantially between 1990 and 2020, and that much of this decline is driven by continuing industry leaders reducing the number of products they produce. Taken together, these findings show that leading firms increasingly grow by expanding within a narrower set of products rather than by producing a broader range of products. This raises the question of what drives this shift in the margin of firm growth and how it relates to the rise in industry concentration over the same period.

3 An oligopolistic competition model of multi-product firms

This section develops an oligopolistic model of multi-product firms to study how improvements in scalable production technologies can lead firms to become larger while producing fewer products. In this model, firms make two choices: how many product markets to enter (product scope), and how much to invest in fixed-cost, scalable technology for each product. Investment lowers product-level marginal costs through expenditures that are fixed with respect to output. Its implementation across products is subject to a common convex firm-level cost, capturing internal resource constraints.

I first present the model and characterize how technological progress changes firms' investment and product scope decisions, and their implications for industry concentration. I then provide supporting empirical evidence connecting the model's predictions to the stylized facts documented earlier.

3.1 Preferences

Households have Cobb-Douglas preferences over industrial output $\{Q^k\}_{k=1}^K$ from K industries indexed by $k \in \{1, \cdots, K\}$. Aggregate utility is given by

$$\log U(Q) = \sum_{k=1}^{K} \alpha^k \log Q^k$$

where $\{\alpha^k\}_{k=1}^K$ are industry expenditure shares that satisfy $\sum_{k=1}^K \alpha^k = 1$ and determine the share of household income spent on consumption across industries.

Within each industry k, production is organized through a two-tier CES structure. One nest aggregates firm-level varieties within each product market, and the other aggregates product-market composites within the industry. The industry contains a continuum of product markets indexed by $j \in [0,1]$. In each product market j, a finite, endogenously determined set of active firms $\mathcal{M}_j^k$ supplies horizontally differentiated varieties. At the lower nest, a competitive product-market assembler combines the horizontally differentiated varieties supplied by these firms into a market-level composite q_j^k. At the upper stage, a competitive final-good producer aggregates the continuum of market-level composites $\{q_j^k\}_{j\in[0,1]}$ into industry output Q^k. A multiproduct firm may participate in several distinct product markets, supplying one differentiated variety in each market it enters. Thus, the demand satisfies,

$$Q^k = \left(\int_0^1 (q_j^k)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1 \tag{3}$$

$$q_j^k = \left(\sum_{i \in \mathcal{M}_j^k} (q_{ij}^k)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \qquad \eta > \sigma > 1 \tag{4}$$

where $\sigma > 1$ is the elasticity of substitution across product markets. q_{ij}^k is the quantity of firm i's variety in market (j,k), and $\eta > 1$ is the elasticity of substitution across varieties within a product market. I assume that $\eta > \sigma > 1$, so that firm-level varieties within a product market are closer substitutes than products within an industry.

With this demand structure, consumer expenditure on product market j in industry k is given by

$$r_j^k = p_j^k q_j^k = S_j^k \alpha^k Y, \qquad \text{with } S_j^k = \left(\frac{p_j^k}{P^k} \right)^{1-\sigma} \tag{5}$$

where p_j^k is the price index of product j in industry k, S_j^k is the within-industry expenditure share of product market j, and Y is aggregate income in the economy. The industrial price index P^k satisfies

$$P^k = \left[\int_0^1 (p_j^k)^{1-\sigma} dj \right]^{\frac{1}{1-\sigma}}. \tag{6}$$

The product market level price index satisfies

$$p_j^k = \left(\sum_{i \in \mathcal{M}_j^k} (p_{ij}^k)^{1-\eta} \right)^{\frac{1}{1-\eta}} \tag{7}$$

where p_{ij}^k is the price charged by firm i in product market (j,k).

Firm-level demand and expenditure within product market (j,k) are therefore given by

$$q_{ij}^k = q_j^k \left(\frac{p_{ij}^k}{p_j^k} \right)^{-\eta}, \tag{8}$$

$$r_{ij}^k = p_{ij}^k q_{ij}^k = s_{ij}^k r_j^k = s_{ij}^k S_j^k \alpha^k Y, \qquad \text{with } s_{ij}^k = \left(\frac{p_{ij}^k}{p_j^k} \right)^{1-\eta} \tag{9}$$

where s_{ij}^k is the expenditure share of firm i within product market (j,k).

Two features of this nested-CES structure distinguish our model from standard models of multiproduct firms. First, the model abstracts from cannibalization across a firm's own products and instead focuses on the supply-side forces shaping firms' scope and scale decisions. In classical multiproduct oligopoly, a firm's varieties compete with one another. In Eckel and Neary (2010), for example, the firm internalizes this demand-side cannibalization when choosing its product portfolio, and cannibalization itself shifts activity toward the firm's core products. In our setting, firms operate a finite number of lines drawn from a continuum of product markets, so the probability that two lines belonging to the same firm appear in the same product market is zero. Products supplied by the same firm therefore do not compete directly in demand. This abstraction is consistent with our empirical definition of a product. Our products are broad product categories rather than individual barcodes or narrowly differentiated varieties. For example, "application software" and "operating systems" are treated as distinct products. At this level of aggregation, modeling a firm's products as direct substitutes within the same demand nest is less natural.

Second, the structure allows a firm to charge different markups across its products. In Hottman et al. (2016), products are nested within firms, so a firm charges a common markup across its product portfolio. I instead organize competition at the product-market level: each product market contains multiple differentiated suppliers, and a multiproduct firm may participate in several distinct markets. This formulation is closer to the product-market structure used in the endogenous-growth literature, but differs from quality-ladder models in which a single producer serves each product market and markups are determined by limit pricing, as in Grossman and Helpman (1991) and Peters (2020). Here, multiple firms can coexist within a product market, and a firm's markup on each product depends on its position in that market. I characterize this pricing decision later.

3.2 Production Technology

Each industry k is populated by an exogenous mass N^k of firms. Firm i in industry k produces q_{ij}^k units in product market (j,k) using labor as the only variable input. The production technology is linear, with labor as the only input,

$$q_{ij}^k = z_{ij}^k l_{ij}^k,$$

where productivity satisfies

$$\log z_{ij}^k = \log A_i^k + \log a_{ij}^k + g\left(u_{ij}^k\right), \quad u_{ij}^k \equiv \psi x_{ij}^k \tag{10}$$

Productivity has three components. The first is firm-level process efficiency A_i^k, which applies to every product supplied by the firm. Firm efficiencies are drawn independently from a Pareto distribution with shape parameter ϑ and lower bound one. I assume that $\vartheta > \eta - 1$ following Chaney (2008).

The second component is product-specific fundamental productivity a_{ij}^k. Following Eckel and Neary (2010), I assume that the product-specific productivity is characterized by a core competence structure, and product opportunities are ranked within a firm. If product j is the product at rank $d_{ij}^k \in \{0, 1, 2, ..., \lambda_i^k - 1\}$ inside a firm i, its fundamental productivity is

$$a_{ij}^k = \delta^{d_{ij}^k}, \qquad \delta \in (0, 1). \tag{11}$$

with $d_{ij}^k = 0$ corresponding to the firm's core product. The length of the list, λ_i^k, is the firm's product-scope choice, described in Section 3.3.3. Thus, product-specific fundamental productivity declines with distance from the core. The parameter δ governs the rate of the decay.

The third component, $g(u_{ij}^k)$, captures the productivity gain from fixed-cost technology investment. Firm i invests $x_{ij}^k \geq 0$ in product j and obtains effective stock

$$u_{ij}^k \equiv \psi x_{ij}^k.$$

I interpret u_{ij}^k as the productivity-relevant capability embodied in the software, data, logistics, and communication systems used in the production of product j. These technologies facilitate production by improving coordination, information processing, and production efficiency, thereby lowering the variable cost of producing each unit of output. They are scalable in the sense that, once installed, the same capability can be applied to a larger volume of production without being depleted. The parameter $\psi > 0$ measures the effectiveness of fixed-cost technology investment.

I assume that $g : \mathbb{R}_+ \to \mathbb{R}_+$ is increasing and concave, with $g(0) = 0$, $g'(u) > 0$, $g''(u) < 0$, and $g'(u) \to 0$ as $u \to \infty$. Thus, product-specific investment raises log productivity but exhibits diminishing returns.

With the wage normalized to one, marginal cost for firm i in product market (j,k) is

$$c_{ij}^k = \frac{1}{z_{ij}^k} = \frac{1}{A_i^k \delta^{d_{ij}^k} \exp\left[g\left(\psi x_{ij}^k\right)\right]}. \tag{12}$$

so that marginal cost is decreasing in firm efficiency, increasing in distance from the core through $\delta^{-d_{ij}^k} \geq 1$, and decreasing in the product-specific investment gain g.

3.3 Market structure and firm decisions

3.3.1 Timing

Firms choose their product portfolios, investment, and pricing strategies in two stages.

Stage 1. In Stage 1, firm i observes its own process efficiency A_i^k and chooses product scope $\lambda_i^k \in \Lambda^k \equiv \{0, 1, ..., \bar{\lambda}\}$. Product scope is the number of product opportunities the firm chooses to develop. Developing each product costs F, so the total product development cost is $F\lambda_i^k$, and $\lambda_i^k = 0$ denotes non-production. The product development costs are incurred before product identities, competitor types, and competitors' portfolios are known.

Random product-market draws. Conditional on λ_i^k, firm i receives an ordered list of λ_i^k distinct product-market opportunities. I refer to this assignment mechanism as random anonymous placement. Product identities are drawn uniformly without replacement from a continuum of product markets $j \in [0,1]$. The first draw is the firm's core line, the second draw has rank one, and so forth. The product at rank d has fundamental productivity δ^d, as defined in equation (11). Every realized opportunity is operated in Stage 2 as there is no fixed operation cost. Thus, λ_i determines how many markets the firm enters, while the random draws determine where it meets competitors[5]. Random anonymous placement rules out strategic portfolio selection. Firms choose product scope before learning product identities and cannot direct entry toward markets with particular competitors or competitive conditions. The realized competition network is therefore

[5]The continuum economy is the large-market limit of economies with J product markets and approximately $N^k J$ firms, where each firm's scope remains bounded as $J \to \infty$. This scaling preserves finite competition in a typical product market. Given a population distribution of G^k of the first-stage states (A, λ), the number of suppliers in a typical product market converges to a Poisson random variable with mean

$$\bar{n}^k = N^k \mathbb{E}_{G^k}[\lambda] \tag{13}$$

Appendix A gives the finite-economy construction and the corresponding marked local-network limit.

generated by random overlap in firms' product portfolios rather than by strategic targeting.

Stage 2. At the beginning of Stage 2, the realized product portfolios, product ranks, and firm efficiencies are publicly observed. Firms then set prices and investment in fixed-cost technologies across their product lines.

I solve the model by backward induction.

3.3.2 Stage 2: Product-Market Competition and Product-level Investment

At the beginning of Stage 2, product-market placement and productivity draws are realized. Firms' process efficiencies, realized portfolios, and product ranks are publicly observed. The public information in industry k is summarized by

$$\mathcal{H}^k \equiv \left\{ \left(A_i^k, \mathcal{J}_i^k, \{a_{ij}^k\}_{j \in \mathcal{J}_i^k} \right) \right\}_{i \in \mathcal{I}^k} \tag{14}$$

where $\mathcal{I}^k$ denote the set of firms in industry k, $\mathcal{J}_i^k$ is firm i's portfolio of product markets, and a_{ij}^k is its product-specific fundamental productivity in market j[6].

In this stage, the key decisions for firm i are the price charged in each product market and its product-specific investment across products in its portfolio.

Firm problem. Given competitors' prices and investment decisions, firm i chooses $\{p_{ij}^k, x_{ij}^k\}_{j \in \mathcal{J}_i^k}$ to maximize operating profit net of technology investment costs. All firms compete in price à la Bertrand. Specifically, firm i solves the following problem.

$$V(A_i^k, \omega_i^k; P^k, Y) = \max_{\{p_{ij}^k, x_{ij}^k\}_{j \in \mathcal{J}_i^k}} \left\{ \sum_{j \in \mathcal{J}_i^k} \left(p_{ij}^k - c_{ij}^k \right) q_{ij}^k - C\left(\sum_{j \in \mathcal{J}_i^k} x_{ij}^k \right) \right\} \tag{15}$$

subject to the marginal cost given by equation (12), demand (8), and total investment cost

$$C\left(\sum_{j \in \mathcal{J}_i^k} x_{ij}^k \right) = \phi_1 \left(\sum_{j \in \mathcal{J}_i^k} x_{ij}^k \right)^{\gamma}, \qquad x_{ij}^k \geq 0, \phi_1 > 0, \gamma > 1$$

[6] Because $a_{ij}^k = \delta^{d_{ij}^k}$, recording a_{ij}^k is equivalent to recording the product's rank d_{ij}^k. Product scope need not enter θ_i^k separately, because $\lambda_i^k = |\mathcal{J}_i^k|$ once the realized portfolio is observed.

Firms are atomistic in each industry since they only operate in a finite number of product markets in an industry with a continuum of products. Therefore, they take the industry-level price index P^k and aggregate income Y as given, while internalizing the effect of their own price on the product-level price index p_j^k.

Here, firms incur a total cost of $C\left(\sum_{j\in\mathcal{J}_i^k} x_{ij}^k\right)$ on their product-level technology investments. The total resource cost is convex in investment across the firm's portfolio, capturing internal resource constraints that link investment decisions across product lines. The parameter γ governs how rapidly the marginal cost of investment rises as the firm expands its total fixed-cost technology investment. Convexity ($\gamma > 1$) is the model's central organizational friction: holding investment elsewhere fixed, greater investment in one product raises the marginal cost of technological upgrading throughout the portfolio. Products within the firm therefore compete for investment resources, so the cost of investing in a given product depends on investment elsewhere in the firm. When $\gamma = 1$, investment costs are separable across products, and this direct cost linkage disappears.

Two features of the firm's problem are worth emphasizing. First, product-level technology investment is endogenous. By choosing how much to invest in fixed-cost technology for each product, the firm endogenously determines the marginal cost at which that product is produced. Technology investment is therefore an additional strategic margin: firms compete not only through prices, but also through investment decisions that shape their production costs. This differs from models in which productivity or marginal cost is an exogenous state of the pricing problem.[7]

Second, the firm-level convex cost of investment makes technology choices nonseparable across products and has two implications. Within the firm, products compete for investment resources. Increasing investment in one product without reducing it elsewhere requires an increase in total firm investment, which raises the marginal cost of investment across the portfolio. A higher return to investment in one product can therefore induce the firm to expand investment there while reducing investment in other products, potentially increasing their marginal production costs.

In the notation of the firm problem (15), although the entire realization $\mathcal{H}^k$ as defined in equation (14) is observed, only a subset of it is relevant for a focal firm i. The relevant

[7] In Bernard et al. (2010), for example, product participation is endogenous, but firm productivity and firm-product appeal are exogenous; conditional on those states, the price of each product is chosen separately. Similarly, in oligopoly models such as Atkeson and Burstein (2008) and Gaubert and Itskhoki (2021), firms compete given their productivity and cost realizations rather than choosing product-level marginal costs through endogenous technology investment.

environment is summarized by ω_i. Formally, for a focal firm i, define its connected set by

$$\mathcal{C}_i^k \equiv \left\{ \ell \in \mathcal{I}^k : \begin{array}{l} \text{there exist } R \in \mathbb{N}_0 \text{ and firms } i_0, \ldots, i_R, \text{ such that} \\ i_0 = i, \quad i_R = \ell, \quad \mathcal{J}_{i,r-1}^k \cap \mathcal{J}_{i,r}^k \neq \emptyset, \text{ for every } r = 1, \ldots, R \end{array} \right\}.$$

Thus, $\mathcal{C}_i^k$ contains firm i, its direct competitors, the competitors of those firms in their other product markets, and all firms connected to i through a finite sequence of shared product markets.

Firm i's payoff-relevant environment is the restriction of the industry realization to its connected set $\mathcal{C}_i^k$. Keeping the focal firm's own state A_i^k separate, define its payoff-relevant environment as

$$\omega_i^k \equiv \left(\mathcal{J}_i^k, \{a_{ij}^k\}_{j \in \mathcal{J}_i^k}, \left\{ \left(A_\ell^k, \mathcal{J}_\ell^k, \{a_{\ell j}^k\}_{j \in \mathcal{J}_\ell^k} \right) \right\}_{\ell \in \mathcal{C}_i^k / \{i\}} \right). \tag{16}$$

where ω_i^k contains its realized portfolio and the characteristics and portfolios of the other firms connected to it through product-market competition. Conditional on P^k and Y, firms outside $\mathcal{C}_i^k$ cannot affect firm i through either product-market prices or a chain of investment-allocation responses.

3.3.3 Stage 1: Product Scope Choice

At the beginning of Stage 1, firm i observes its process efficiency A_i^k. It chooses a finite number product scope $\lambda_i^k \in \Lambda^k \equiv \{0, 1, ..., \lambda^k\}$ by solving the following Stage-1 problem,

$$W\left(A_i^k; \Gamma^k, P^k, Y\right) = \max_{\lambda_i^k \in \Lambda^k} \left\{ \mathbb{E}_{\omega \sim \Gamma^k(\lambda)} \left[V\left(A_i^k, \omega; P^k, Y\right) \right] - F\lambda_i^k \right\}. \tag{17}$$

where $V(A_i^k, \omega; P^k, Y)$ is the firm profit in Stage-2 given realization of payoff-relevant environment ω, as defined in (16). This environment is realized only after the firm has chosen its product scope. $F\lambda_i^k$ denotes the fixed cost of developing λ_i^k product opportunities.

The expectation in (17) is taken over the distribution of firm's payoff-relevant environment. The distribution of the payoff-relevant environment ω when firm i chooses scope λ is written as

$$\omega_i^k(\lambda) \sim \Gamma^k(\lambda), \tag{18}$$

I collect these scope-contingent distributions in

$$\Gamma^k \equiv \left\{\Gamma^k(\lambda)\right\}_{\lambda\in\Lambda^k}.$$

An individual firm takes the entire family Γ^k as given. Its scope choice selects the distribution $\Gamma^k(\lambda)$ relevant for its own future environment, but does not alter that distribution since each firm is atomistic in the industry.

The family Γ^k deserves more discussion here. It is a general equilibrium object determined endogenously. Let G^k denote the cross-sectional distribution of firm characteristics (A_i^k, λ_i^k) generated by firms' optimal Stage-1 choices. Given G^k, random product-market placement determines the distribution of firms, product ranks, and portfolios in the competitive environment surrounding a focal firm, and hence determines $\Gamma^k(\lambda)$ for every λ. In equilibrium, the distributions used by firms in (17) coincide with those generated by firms' choices and the random placement process.

3.4 Equilibrium

We close the decentralized economy by imposing labor-market clearing. The representative household supplies $\bar{L}$ units of labor inelastically, and the wage is normalized to one. Labor is used for variable production, technology investment, and product development. Let $X_i^k \equiv \sum_{j\in J_i^k} x_{ij}^k$ denote firm i's total technology investment. Labor-market clearing requires aggregate labor demand to equal aggregate labor supply:

$$\bar{L} = \sum_{k=1}^{K} N^k \int \left\{ F\lambda + \mathbb{E}_{\omega\sim\Gamma^k(\lambda)} \left[\sum_{j\in J_i^k} l_{ij}^k + C\left(X_i^k\right) \right] \right\} dG^k(A,\lambda). \tag{19}$$

The three terms on the right-hand side are labor used for product development, variable production, and technology investment, respectively. All Stage-2 objects inside the expectation are evaluated at firm productivity A and the realized competitive environment ω.

A symmetric Bayesian equilibrium consists, for each industry k, of a scope policy $\lambda^k(A)$, an induced distribution of firm characteristics G^k, a family of scope-contingent competitive environments $\{\Gamma^k(\lambda)\}_{\lambda\in\Lambda^k}$, Stage-2 prices and technology investments $\{p_{ij}^k, x_{ij}^k\}$ and the associated quantities $\{q_{ij}^k\}$, and an industry price index P^k, together with aggregate income Y, such that: firms' scope choices solve (17); for every realized competitive environment, each firm's price and technology-investment choices solve its Stage-2 prob-

lem, taking competitors' choices and the aggregate objects (P^k, Y) as given; G^k and Γ^k are generated by firms' optimal policies and the random product-market assignment process; demand, production, and price indices are mutually consistent; and the aggregate labor market clears according to (19).

Equilibrium is therefore a fixed point between firms' policies and the competitive environments they generate. Scope policies determine the cross-sectional distribution of firms and product-market overlap. These distributions determine firms' Stage-2 prices and investments, while aggregate income adjusts so that total labor demand equals the fixed labor supply. Appendix Section C.1 provides the complete system of equilibrium conditions.

3.5 Existence and Uniqueness of the Equilibrium

Below, I provide theorems for the existence and uniqueness of both the Stage-2 equilibrium and the full equilibrium. (Pending)

4 Model Characterization

This section first characterizes the optimality conditions governing firms' pricing, product-level investment, and product scope choices. I then examine how these choices respond to improvements in technology efficiency. Finally, I study how these firm-level responses affect industry concentration, markups, and aggregate productivity.

I proceed in two steps. Section 4.1 studies firms' investment and product scope choices in a monopolistic-competition benchmark. I assume that each firm-product variety is infinitesimal within its product market and allow product scope to be continuous. These simplifications preserve the core–periphery productivity structure and endogenous technology investment, while removing variable markups, strategic interactions, and discrete changes in product scope. I use this benchmark to characterize firms' responses to a common improvement in technology efficiency and derive an exact marginal condition for product scope. The analysis shows how better fixed-cost technology can allow firms to grow without expanding their product portfolios.

Section 4.2 returns to the full oligopolistic model with a finite number of suppliers in each product market and discrete product scope choices. I characterize how firms respond to improvements in technology efficiency and study the implications for industry concentration, markups, and aggregate productivity.

4.1 Monopolistic competition benchmark

In this benchmark, I take the expected number of producers per product market, defined in equation (13), approach infinity. Each firm is therefore infinitesimal not only within its industry but also within each product market. As a result, each firm faces a constant demand elasticity η and charges a constant markup $\mu = \eta/(\eta - 1)$ over marginal cost. In this setting, firm i chooses product scope $\lambda_i \in [0, 1]$, interpreted as the measure of ranked product opportunities it develops. Under anonymous product-market placement, these opportunities are assigned to a set of distinct product markets of measure λ_i. Because placement is independent of firm characteristics and firms are atomistic, all product markets have the same distribution of supplier types and product ranks.

Suppress the industry superscript and write $E = \alpha Y$, $\zeta = -\log \delta$. Firm i chooses $\lambda_i \in [0, 1]$ and $x_i(r) \geq 0$ for ranks $r \in [0, \lambda_i]$, with $r = 0$ the core. Effective investment is $u_i(r) = \psi x_i(r)$ and productivity is $z_i(r) = A_i e^{-\zeta r} e^{g(u_i(r))}$. Operating profit and the common competition index are

$$\pi_i(r) = \frac{E}{\eta \Omega} z_i(r)^{\eta - 1}, \qquad \Omega = N \int \int_0^{\lambda(A)} z_A(r)^{\eta - 1} \, dr \, dG(A), \tag{20}$$

where G is the distribution of firm efficiency and N is the mass of potential firms. (Notes: Ω measures the average productivity of the industry). A larger Ω lowers profitability at given E. Writing $X_i = \int_0^{\lambda_i} x_i(r) dr$ and $C(X_i) = \phi_1 X_i^{\gamma}$, the firm solves

$$\max_{\lambda_i \in [0,1],\, x_i(\cdot) \geq 0} \left\{ \int_0^{\lambda_i} [\pi_i(r) - F] \, dr - C(X_i) \right\} \tag{21}$$

We characterize the firm optimal choices in investment and product scope below.

Optimal product-level investment. Consider the firm's product-specific fixed-cost technology investment choice. Let $\nu_i^k = C'\left(X_i^k\right) = \phi_1 \gamma (X_i^k)^{\gamma - 1}$ denote the marginal investment cost for firm i. Since $\frac{\partial ci(r)}{\partial x_i(r)} = -\psi c_i(r) g'\left(u_i(r)\right)$, the investment optimality condition is

$$\psi q_i(r) c_i(r) g'\left(u_i(r)\right) = \psi l_i(r) g'(u_i(r)) \leq \nu_i, \qquad u_i(r) = \psi x_i(r), \quad x_i(r) \geq 0. \tag{22}$$

with equality whenever $x_i(r) > 0$.

Two forces govern this investment condition. First, the benefit from investment depends on the product's production base. The term $q_i(r) c_i(r)$ is the product's total variable

cost. Because labor is the only variable input and the wage is normalized to one, this term also equals production employment, $l_i(r)$. The left-hand side of equation (22) therefore measures the labor savings generated by additional product-level investment.

An important feature of technology investment is that it is scale-biased. Holding the marginal investment cost fixed, investment is more valuable for a product with a larger production base. A producer with higher fundamental productivity, $A_i\delta^r$, operates at a larger scale and therefore have stronger incentives to invest. Higher investment further lowers their marginal production costs and expands their production scale, reinforcing their initial advantage. Fixed-cost technology investment can therefore amplify initial productivity differences across producers. In this benchmark, conditional on the same product scope, investment at a given rank increases with firm productivity A_i (Notes: also conditional on the same marginal cost. So just say that, firms with higher productivity for the same rank product have higher incentive to invest?).

The second force operates on the cost side. Optimal investment in each product depends on the common marginal investment cost, ν_i, which rises with the firm's total investment when $\gamma > 1$. Greater investment in one product raises the marginal cost of investment throughout the portfolio, inducing the firm to reduce investment elsewhere. In this sense, products within the firm compete for investment resources through the common marginal cost. The firm allocates investment by equating marginal returns across products receiving positive investment. Consequently, investment in a given product depends not only on conditions in its own market, but also on the firm's investment opportunities elsewhere in its portfolio.

Optimal product scope. Consider next the firm's product scope choice.[8] At an interior optimum, the marginal product satisfies

$$\pi_i(\lambda_i) = F + \nu_i x_i(\lambda_i) \tag{23}$$

Let $\frac{\partial X_i}{\partial \lambda_i}$ denote the change in total investment when scope expands and investment on existing products is re-optimized. Then

$$\nu_i x_i(\lambda_i) = \underbrace{\nu_i \frac{\partial X_i}{\partial \lambda_i}}_{\text{additional investment expenditure}} + \underbrace{\nu_i \left(x_i(\lambda_i) - \frac{\partial X_i}{\partial \lambda_i} \right)}_{\text{profit forgone through investment crowd-out}} \tag{24}$$

[8]The firm chooses scope before its product assignments are realized. Under the monopolistic competition benchmark, however, every assignment delivers the same competitive environment. The scope problem is therefore deterministic.

These conditions make the breadth–depth trade-off explicit. The marginal product's operating profit must cover three costs: the development cost F, additional technology investment expenditure, and operating profits forgone on existing products due to investment crowding out. The third component represents the opportunity cost of investing in the new product. Expanding scope therefore comes partly at the expense of technological upgrading in existing products, raising their unit production costs and weakening their competitive positions. Conversely, narrowing scope relaxes this pressure and allows firms to invest more in their retained products. When investment costs are separable across products, as in the case of linear firm-level investment costs, this internal crowding-out channel disappears. Firms can then expand through product scope without directly crowding out investment in existing products, as in models emphasizing product expansion as a margin of firm growth (Bernard et al. 2011; Klette and Kortum 2004).

Substituting equation (20) into equation (23) gives the implicit scope characterization

$$\lambda_i = \frac{1}{\zeta}\left[\log A_i + g(u_i^\lambda) + \frac{1}{\eta - 1}\log\left(\frac{E}{\eta\Omega(F + \nu_i x_i^\lambda)}\right)\right]. \tag{25}$$

Higher fundamental productivity favors broader scope, while greater internal congestion limits expansion. The following lemma establishes the net cross-sectional relationship.

Lemma 1 (Scale, scope, and internal congestion)**.** *Suppose g is continuous at zero with $g(0) = 0$, twice continuously differentiable for $u > 0$, and satisfies $g'(u) > 0$ and $g''(u) + (\eta - 1)[g'(u)]^2 < 0$ for all $u > 0$. Holding E/Ω and ψ fixed, firms with higher firm-level productivity A_i have greater total revenue and production employment, choose broader product scope, and make greater total investment. When $\gamma > 1$, they also face a higher marginal investment cost ν_i.*

Proof: See Appendix C.2.

Response to technology shocks. Having characterized firms' optimal choices and their implications for the firm-size distribution, I next examine how improvements in technology efficiency affect these choices and the resulting distribution. Specifically, I consider a small, common increase in technology efficiency ψ, holding industry expenditure αY fixed. Both product scope and investment adjust, and the common price index P responds to these choices.

The shock affects both competition within product markets and the value of retaining peripheral products. With constant markups, the sales share of a continuing supplier i in

product market j responds according to

$$\frac{d\log s_{ij}}{d\log\psi} = (\eta-1)\left(\frac{d\log z_{ij}}{d\log\psi} + \frac{d\log P}{d\log\psi}\right), \tag{26}$$

and

$$\frac{d\log z_{ij}}{d\log\psi} = \underbrace{\frac{\kappa(u_{ij})}{1-(\eta-1)\kappa(u_{ij})}}_{\text{technology responsiveness}} \underbrace{\left[1+(\eta-1)\frac{d\log P}{d\log\psi} - \frac{d\log\nu_i}{d\log\psi}\right]}_{\substack{\text{investment incentive net of} \\ \text{competition and internal congestion}}}. \tag{27}$$

Equation (26) compares the supplier's productivity gain with the decline in the common price index P: suppliers whose prices fall faster than the index gain market share. Equation (27) states that how much productivity gaps change depends on the shape of the technology, initial investment intensity, and the response of the parent's marginal investment cost. The curvature restriction in Lemma 1 makes the first factor positive. If κ increases with effective investment (explain what is kappa here, economically, in words and short), it is larger for products that are initially more investment-intensive. The second factor is common within a firm but differs across parents: a faster increase in ν_i dampens all of a firm's productivity gains.

The change in technology also change the firms' scope choices. Differentiating the marginal-product condition (23), and applying the envelop theorem gives

$$(-\log\delta)\frac{d\lambda_i}{d\log\psi} = \underbrace{\left[u_i^\lambda g'(u_i^\lambda) + \frac{d\log P}{d\log\psi}\right]}_{\substack{\text{within-product-market competition:} \\ \text{own gain relative to competitors}}} - \underbrace{u_i^\lambda g'(u_i^\lambda)\frac{d\log\nu_i}{d\log\psi}}_{\text{internal investment congestion}}. \tag{28}$$

Here, $u_i^\lambda = \psi x_i(\lambda_i)$ is effective investment at the margin. The first bracket combines the direct effect of better technology on the marginal product's optimized contribution with the change in market prices, which reflect the changes in levels of compeition within the product market. The second captures the change in its internal opportunity cost: higher total investment raises ν_i, making peripheral investment more expensive.

Formally, the propositition below formalizes the idea that, they deliver a joint pattern of refocusing, leading-firm specialization, and reallocation toward productive suppliers consistent with the paper's motivating evidence.

Proposition 1 (Scale-biased upgrading and product scope)**.** *Consider a small common increase in ψ, holding industry expenditure fixed. Maintain the conditions of Lemma 1, and suppose $\kappa(u) = [g'(u)]^2/[-g''(u)]$ is strictly increasing, $ug'(u)$ is non-decreasing, and $\gamma > 1$. (and*

other assumptions?)

(a) Product-market reallocation. In each product market, suppliers above any sufficiently high pre-shock fundamental-productivity cutoff in $A_i\delta^{r_{ij}}$ increase their combined market share.

(b) Within-firm refocusing. Among retained products, those with higher fundamental productivity obtain larger proportional productivity gains and faster revenue growth.

(c) Leading-firm specialization. There exists a finite A^* such that firms with $A_i > A^*$ increase total investment and sales while reducing scope. Their marginal investment cost rises, and the combined industry sales share of any fixed group $A_i \geq A_0 > A^*$ increases.

Proof. See Appendix C.3.

The proposition connects product-market reallocation to the internal allocation of investment. More productive firms invest more overall and face a higher marginal investment cost, leaving their own marginal products lightly upgraded. Following the shock, sufficiently productive firms increase investment and its marginal cost further, while these peripheral products receive little direct technological gain. The decline in the market price index and the higher internal cost lower the contribution of the previously marginal product below F. The firm withdraws it; its new marginal product again contributes F.

Leading firms therefore grow through greater scale in a narrower portfolio, not by expanding product scope. Their total sales rise after accounting for withdrawn products, so their industry sales share increases. Within product markets, the gain is established for high-fundamental-productivity suppliers as a group, allowing their parents' investment costs to differ and adjust; it does not require every retained product of a leading firm to gain share.

4.2 Firm choices under oligopolistic competition

We now return to the full oligopoly model. This restores strategic competition among firms within product markets. We characterize the optimal pricing, decisions of the firm in this setting, while deferring more detailed discussion on the optimal scope and inestment to the Appendix C.4.

Optimal pricing. In Bertrand-Nash equilibrium, firm i's price in product market (j,k) satisfies

$$p_{ij}^k = \mu_{ij}^k c_{ij}^k, \qquad \mu_{ij}^k = \frac{\varepsilon_{ij}^k}{\varepsilon_{ij}^k - 1}, \qquad \varepsilon_{ij}^k \equiv \varepsilon(s_{ij}^k) = \sigma s_{ij}^k + \eta(1 - s_{ij}^k). \tag{29}$$

Here, s_{ij}^k denotes firm i's expenditure share in product market (j,k), as defined in equation (9). Under the nested CES demand system, the residual demand elasticity faced by the firm is a sales-share-weighted average of the across-product elasticity σ and the within-product elasticity η, similar in spirit to Gaubert and Itskhoki (2021). Since $\eta > \sigma$, this elasticity falls with the firm's within-product market share s_{ij}^k. As a result, the markup increases with the firm's market share in product market (j,k).

Because each firm supplies only finitely many products in a continuum of product markets, it takes the industry price index P^k as given. Conditional on marginal costs, pricing interactions are therefore confined to individual product markets. The common investment cost is the only direct link across a firm's products in Stage 2.

4.3 Product market outcomes

Holding product portfolios fixed, I next examine how the heterogeneous productivity gains characterized above affect product-market concentration, cost-weighted markups, and production productivity.

For each product market j, let $l_j = \sum_i l_{ij}$ and $\ell_{ij} = l_{ij}/l_j$. Define concentration $H_j = \sum_i s_{ij}^2$, the cost-weighted markup $\mu_j = \sum_i \ell_{ij}\mu_{ij} = [\sum_i s_{ij}/\mu_{ij}]^{-1}$, and production productivity $Z_j = q_j/l_j = \mu_j/p_j$. Substituting the pricing rule into the CES price index yields

$$Z_j = \left[\sum_{i \in \mathcal{M}_j} \left(\frac{\mu_{ij}}{\mu_j}\right)^{1-\eta} z_{ij}^{\eta-1}\right]^{1/(\eta-1)}. \tag{30}$$

This equation highlight the relationship between product-market markup dispersion and productivity. With uniform markups, this reduces to the standard power mean $(\sum_i z_{ij}^{\eta-1})^{1/(\eta-1)}$. Markup dispersion twists the allocation of activities across suppliers, thus reduce the product market productivity. This is the same force as in the standard literature like Edmond et al. (2023). Moreover, the aggregate carries a love-of-variety force: a reduction in the number of suppliers would result in a loss in productivity.

For a common improvement in IT efficiency, product-market concentration and the

cost-weighted markup respond as,

$$\frac{dH_j}{d\log\psi} = \underbrace{2\bar{\varsigma}_j \operatorname{Cov}_{w_j}\left(s_{ij}, G_{ij}\right)}_{\text{reallocation toward larger suppliers}}, \tag{31}$$

$$\frac{d\log\mu_j}{d\log\psi} = \underbrace{(\eta-\sigma)\mu_j\bar{\varsigma}_j \operatorname{Cov}_{w_j}\left(\frac{s_{ij}}{[\eta-(\eta-\sigma)s_{ij}]^2}, G_{ij}\right)}_{\text{individual markup adjustment}} + \underbrace{\mu_j\bar{\varsigma}_j \operatorname{Cov}_{w_j}\left(\frac{1}{\eta-(\eta-\sigma)s_{ij}}, G_{ij}\right)}_{\text{reallocation across suppliers}}, \tag{32}$$

When initially larger suppliers obtain larger productivity gains from IT, they gain market share and product market concentration rises. The cost-weighted markup also rises through two components: individual markup adjustment and expenditure reallocation toward suppliers with higher markups.

Production productivity responds through technological improvement and output reallocation

$$\frac{d\log Z_j}{d\log\psi} = \underbrace{\sum_{i\in\mathcal{M}_j}\ell_{ij}G_{ij}}_{\text{technological improvement}} + \underbrace{\frac{\eta\mu_j\bar{\varsigma}_j}{\eta-1}\operatorname{Cov}_{w_j}\left(\frac{1}{\eta-(\eta-\sigma)s_{ij}}, G_{ij}\right)}_{\text{output reallocation}}. \tag{33}$$

The first term in Equation (33) The first term captures the productivity gains from improved technologies. When these gains are larger for initially larger suppliers, the resulting output reallocation provides an additional productivity gain (Baqaee and Farhi 2020). Higher productivity, concentration, and markups can therefore arise jointly from the same distribution of technological gains.

These responses hold product portfolios fixed. Scope adjustment also changes the set of suppliers and the resources devoted to product development. Product withdrawals remove varieties and can strengthen the positions of remaining suppliers, while lower development costs release resources for production. The quantitative analysis evaluates these participation and resource effects jointly with the investment responses characterized here.

5 Empirical Evidence for the Model Mechanism

I next provide reduced-form empirical evidence consistent with the model's key assumptions and predictions, using information technology (IT) as an example of scalable fixed-cost investment. IT is particularly relevant because its adoption increased rapidly over the same period in which industry concentration rose and leading firms' product scope declined (Lashkari et al. 2024). Examples of fixed-cost IT investments include specialized software embedded in production machinery, machine-tool programming, and automated control systems. These IT investments often require upfront expenditures but can support larger production volumes without proportional increases in costs (Crouzet et al. 2022). These features make IT a natural setting for examining the model's mechanism.

Using industry- and firm-level data, I document two main findings. First, industries with larger increases in IT capital per worker experience greater declines in leading firms' product scope. Second, firm-level evidence is consistent with convex investment costs. Using panel data, I find that increases in firms' IT investment are associated with reductions in their product scope.

5.1 Data

CI Technology Database. I measure firm-level information technology using the Aberdeen CI Technology Database, formerly known as the Harte Hanks CI Technology Database. The database provides establishment-level information on firms' use of information technology from 1998 to 2019. It includes measures of annual expenditures on hardware, software, and communications equipment. The database has been widely used in research examining the role of IT in firm organization and productivity (Bresnahan et al. 2002, Bloom et al. 2012).

The database also reports establishment-level DUNS numbers, which allow me to link its records to the Dun & Bradstreet data. I aggregate these establishment-level records to construct annual firm-level measures of IT capital and IT spending.

5.1.1 IT exposure and top firm product scope: Industry-level evidence

I first use industry-level data to examine whether industries with larger improvements in IT efficiency experience greater declines in the product scope of their top 1% firms. Because improvements in IT efficiency are not directly observed at the industry level, I

follow Michaels et al. (2014) and use growth in IT capital per worker as a proxy for industry exposure to these improvements. The underlying assumption is that improvements in IT efficiency reduce production costs more in industries that rely more heavily on IT for technological reasons. These industries are therefore expected to increase IT adoption more rapidly. Specifically, I measure industry-level IT exposure using the annual log change in IT capital per worker:

$$\Delta\text{Exposure}_{k,t} = \log\left(\frac{\text{IT Capital}_{k,t}}{L_{k,t}}\right) - \log\left(\frac{\text{IT Capital}_{k,t-1}}{L_{k,t-1}}\right), \tag{34}$$

where $\text{IT Capital}_{k,t}$ denotes the KLEMS measure of IT capital input in industry k and year t, and $L_{k,t}$ denotes total industry employment. In the KLEMS data, IT capital input is measured as compensation to hardware and communications equipment and is available beginning in 1997. The KLEMS measure is reported at the two-digit NAICS level. I concord two-digit NAICS industries to two-digit SIC industries following Schaller and DeCelles (2021). Online Appendix Section XX provides additional details on the data construction.

To estimate the relationship between changes in IT exposure and changes in the product scope of top firms across industries, I estimate the following regression:

$$\Delta y_{k,t} = \alpha\Delta\text{Exposure}_{k,t} + X'_{k,t}\beta + \gamma_k + \varepsilon_{k,t}. \tag{35}$$

Here, $\Delta y_{k,t}$ is the change in the product scope of the top 1% of firms in industry k between years $t-1$ and t, and $\Delta\text{Exposure}_{k,t}$ is the change in IT capital per worker over the same period. The vector $X_{k,t}$ contains additional industry-level controls. Industry fixed effects, γ_k, absorb persistent industry-specific differences in changes in product scope, corresponding to industry-specific trends in the level of product scope.

The vector $X_{k,t}$ includes industry-level controls intended to account for other factors that may affect changes in firm product scope. First, I control for initial industry concentration using the industry-level HHI in year t. I also control for the change in industry concentration between $t-1$ and t. Third, I control for the industry-level value-added. Finally, import competition may affect firms' product-scope decisions (Amiti and Heise 2025). I therefore additionally control for the log change in import from China between year $t-1$ and t.

Table 2 reports the estimates of equation (35). Column (1) reports the estimated coefficient α without additional controls, weighting industries by employment in 1997. The relationship is negative: industries with larger increases in IT exposure experience larger

declines in the product scope of their top 1% firms. A 1 log point increase in IT exposure is associated with a reduction of 0.36 in product scope. Column (2) adds the full set of industry-level controls but the import competition shock, and the negative relationship remains. Column (3) further controls for import competition from China. We find that the China shock is not significantly associated with the reduction in the product scope of top 1% manufacturing firms. Across these specifications, the estimated relationship remains negative and the magnitude of the coefficient is broadly stable.

Table 2: Top firm scope and IT exposure: industry-level regression

	(1)	(2)	(3)
Δ Log IT per worker	-0.357***	-0.290**	-0.310***
	(0.137)	(0.108)	(0.103)
Initial log HHI		0.091***	0.094***
		(0.020)	(0.019)
Δ HHI		0.109***	0.099***
		(0.037)	(0.034)
Δ log(VA)		0.241	0.136
		(0.167)	(0.179)
Δ log(China Import)			0.010
			(0.077)
Observations	460	460	460
R-squared	0.071	0.101	0.103

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table report regression results by estimating equation (35). Column (1) reports the estimated coefficient α without additional controls. Column (2) adds the full set of industry-level controls but the import competition shock. Column (3) further controls for the China import shock. We cluster the standard errors at the industry level.

These industry-level patterns suggest that industries with larger increases in IT exposure experience larger reductions in the product scope of their leading firms. This evidence is consistent with the model prediction in Proposition 1.

5.1.2 Convex firm-level investment cost and selective expansion: Firm-level evidence

A key assumption of the model is that product-specific investments in fixed-cost technology are subject to a common convex firm-level cost. As a result, increasing investment in one product raises the marginal cost of investing in other products. Products therefore compete for investment resources within the firm, creating internal congestion on

the supply side. While previous studies have examined competition for resources within firms, I focus on its implications for product scope. Specifically, I use firm-level panel data to examine whether increases in IT investment are associated with firms reducing the number of products they produce.[9]

In this section, I explore firm data and estimate the relationship between changes in firm product scope and firm investment in IT. These estimates would track whether the same firm increase or reduce their product scope when they adopt more IT.

$$\Delta y_{ikt} = \beta \Delta \text{IT}_{ikt} + X'_{ikt}\beta + \alpha_{kt} + \varepsilon_{ik}. \tag{36}$$

The dependent variable, Δy_{ikt}, is the change in firm i's product scope within industry k between years $t-1$ and t. On the right-hand side, ΔIT_{ikt} denotes the log change in firm-level real investment in IT over the same period. The vector X_{ikt} includes controls for firm age, initial product scope, initial employment, and initial IT capital in period $t-1$. These controls account for differences in firms' life-cycle stages and initial size and scope. Industry-by-year fixed effects, α_{kt}, absorb industry-specific changes over time in product-scope dynamics.

Table 3 reports the regression results. Standard errors are clustered at the firm level. Column (1) examines changes in firm product scope using the full sample. Doubling firm-level PCs per worker is associated with a reduction of 0.3 products. Column (2) restricts the sample to firms in the top 1% of the employment distribution. For these firms, doubling PCs per worker is associated with a reduction of 1.5 products. In data, PC intensity among top 1% firms grows by an average of 7.3% per year in our sample. A simple back-of-the-envelope calculation implies that, at this rate, the estimated relationship corresponds to a decline of approximately 3.29 products over 30 years.

A potential concern is endogeneity. As commonly emphasized in the literature, positive unobserved productivity shocks may induce firms to adopt more IT. However, such shocks would also tend to expand firms' product scope, generating an upward bias in the estimated relationship between IT adoption and product scope. Despite this potential positive bias, I find a negative coefficient. This suggests that the negative relationship documented here is unli

[9]Previously, there has been a strand of literature documenting the internal competition across different plants. See, for example, Giroud and Mueller (2015); Kehrig and Vincent (2026).

Table 3: IT Spending, Product Scope, and Employment

	Δ log Scope		Δ log Employment	
	(1) Full	(2) Top 1%	(3) Full	(4) Top 1%
Log real IT investment	-0.006*	-0.014**	0.014	0.037*
	(0.003)	(0.006)	(0.010)	(0.019)
Log employment, t	0.087***	0.093***	-0.481***	-0.515***
	(0.013)	(0.019)	(0.028)	(0.045)
Log real IT capital, $t-1$	-0.006	-0.006	-0.023*	-0.014
	(0.004)	(0.006)	(0.013)	(0.020)
Firm FE	Yes	Yes	Yes	Yes
SIC2 × year FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	911,762	18,353	911,762	18,353

Notes: This table report regression results by estimating equation (36). Standard errors are clustered at the firm level. Standard errors in the parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Model Quantification

I now turn to estimate the model. I first discuss the calibration and structural estimation strategy and then discuss the extent to which the model is able to replicate a set of targeted and untargeted moments.

6.1 Technology Productivity Function

In our baseline calibration I specify a fixed-cost technology productivity function that allows me to bring the model to the data. Specifically, I assume that the endogenous productivity that's coming from the fixed cost investment in equitation (10) as

$$g(x_{ij}) = \log\left(1 + \psi x_{ij}^{\beta}\right), \quad 0 < \beta < 1 \tag{37}$$

Here, x_{ij} denotes fixed cost investment specific to product j. Also, this functional form satisfies the properties specified in Proposition 2. We normalize the efficiency for the fixed cost technology in equation (37) to be 1 in the baseline calibration.

6.2 Calibration

In the baseline calibration, I calibrate the parameters using the 2020 economy. I divide the model's parameters into two groups. The first group includes parameters that govern household preferences. This group consists of commonly-used building blocks from the literature and I rely on existing estimates to calibrate it. The final group includes parameters from the production block, in particular the services production function and the cost of expansion. This block includes new features relative to the existing literature, and so I estimate it using Simulated Method of Moments (henceforth SMM).

6.2.1 Externally Calibrated Parameters

The parameters reported in Table 4 are either calibrated externally or taken from existing estimates in the literature. We set σ, the elasticity of substitution across product markets defined at the eight-digit SIC level, to 2.5, according to Broda and Weinstein (2006). We set the elasticity of substitution within product markets to 5.0, according to Hottman et al. (2016). We draw on different sources for the two elasticities because neither paper provides estimates for both margins at the level of aggregation used in our model.[10] In Appendix XX, we re-estimate the model using alternative values of σ and η as robustness checks.

Table 4: External Calibrated Parameters

Parameter	Interpretation	Source	Value
σ	EoS across product markets	Broda and Weinstein (2006)	2.5
η	EoS within product markets	Hottman et al. (2016)	5.0
N	Mass of potential entrants	External calibrated	30
ψ	Efficiency of production technology	Normalization	1.0

Finally, in the simulation, we set the expected number of firms per product market to 30, based on the mean number of firms per product market in the 2020 Dun & Bradstreet data. We also normalize the efficiency for the fixed cost technology in equation (37) to be 1 in the baseline calibration.

[10] Broda and Weinstein (2006) does not estimate substitution within narrowly defined product markets, while Hottman et al. (2016) does not report an across-product-market elasticity corresponding to our eight-digit SIC definition. Specifically, $\sigma = 2.5$ matches the median elasticity of substitution for disaggregated 3- to 5-digit SITC categories (2.2 to 2.8) and differentiated goods (2.1 to 2.5) estimated by Broda and Weinstein (2006). $\eta = 5.0$ lies between the median cross-firm elasticity (3.9) and median across-barcode elasticity (6.9) within product groups estimated by Hottman et al. (2016), and matches the standard within-market benchmark in quantitative oligopoly models (for example, Atkeson and Burstein (2008)).

6.2.2 Simulated Method of Moments for the Structurally Calibrated Parameters

The remaining six parameters are estimated by indirect inference, matching model-implied moments to their empirical counterparts. These parameters are the shape parameter of the Pareto distribution, ϑ; the rate at which productivity declines along a firm's product ladder, δ; the fixed cost of creating a product, F; and the three parameters governing the productivity and investment cost function, (β, ϕ_1, γ). I collect these parameters in the vector Θ and estimate them jointly using a set of empirical moments.

$$\Theta \equiv \{\vartheta, \delta, F, \beta, \phi_1, \gamma\}$$

I next discuss which moments are most informative about each parameter and describe how key moments are constructed. In practice, however, all moments are jointly determined in equilibrium by the full parameter vector.

I use employment data from Dun & Bradstreet to discipline the parameters $(\vartheta, \delta, \beta)$. The Pareto shape parameter ϑ governs dispersion in fundamental productivity and, in turn, employment across firms. To discipline ϑ, I estimate the slope of the upper tail of the firm-employment distribution. Following Gabaix and Ibragimov (2011), I rank firms by employment within each two-digit SIC industry and estimate

$$\log\left(r_{ik} - \frac{1}{2}\right) = \alpha_k - \zeta_L \log L_{ik} + \varepsilon_{ik}, \tag{38}$$

where r_{ik} is firm i's employment rank within industry k, L_{ik} is its total employment, and α_k denotes industry fixed effects. I estimate the regression using the top 10% of firms by employment within each industry and target the estimated tail exponent $\widehat{\zeta}_L$.

The employment gap between a firm's core and second-ranked products are informative about parameters δ and β. Specifically, the parameter δ governs how quickly productivity declines from a firm's core product to lower-ranked products. I therefore discipline δ using the average employment gap between the core and second-ranked products among multiproduct firms that are at the bottom 50% of the firm size distribution. To discipline the returns to technology investment, I use differences in the allocation of employment across products among firms of different sizes. The parameter β governs the curvature of productivity gains from product-level investment. I target the difference in the core-to-second product employment gap between firms in the top decile and the bottom half of the employment distribution. This moment captures how the relative scale of a firm's core product varies with firm size and helps discipline the returns to technology investment. Together, these two moments inform the allocation of employment across

products within firms.

Specifically, I estimate the following regression:

$$\log L_{i(1)} - \log L_{i(2)} = \alpha + b\, \mathbb{1}(\text{Top } 10\%_i) + \varepsilon_i, \tag{39}$$

where $L_{i(1)}$ and $L_{i(2)}$ denote employment in firm i's largest and second-largest products, respectively. The indicator $\mathbb{1}(\text{Top } 10\%_i)$ equals one if firm i belongs to the top decile of the employment distribution within its industry. The coefficient b measures the difference in the average log employment gap between firms in the top decile and the remaining firms.

Table 5 reports the regression results of equation (39) using different weights. Column (1) reports ther estuls without any weights, while column (2) report results with firms observations weighted by its employment size. We see that the estimate for the α is consistent, where the core to the second product's gap in employmet size is 1 log points. The difference in the core to second gap among the top 10% firms and the bottom 50% multi-product firms are 0.96 log points when there is no weights, and is 1.36 log points when ther eis weights. We use the estimation results form the first colum nas our baseline moment targets.

Table 5: Regression: Core-to-second product employment gap

	(1) Unweighted	(2) Employment weighted
Top 10% employment indicator	0.962***	1.359***
	(0.036)	(0.082)
Constant (bottom-half mean)	1.002***	1.040***
	(0.021)	(0.024)
Observations	3,955	3,955
Top-decile observations	2,826	2,826
Bottom-half observations	1,129	1,129
Firm clusters	3,869	3,869
R-squared	0.092	0.006

Notes: This table report regression results by estimating equation (39). The dependent variable is $\log L_{i(1)} - \log L_{i(2)}$. Column (1) report unweighted regression results. Column (2) uses total firm-level employment as analytic weights. Standard errors in the parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, I use firm-level revenue and variable-cost data from Compustat to discipline the investment cost parameters (ϕ_1, γ). The parameter ϕ_1 governs the scale of technology investment costs, while γ determines how rapidly marginal investment costs rise with total firm investment. I target the average revenue-to-variable-cost ratio, R_i/VC_i, and how this ratio varies with firm size.

Specifically, I estimate

$$\log\left(\frac{R_i}{VC_i}\right) = \alpha_k + b_{RVC}\log R_i + \varepsilon_i, \tag{40}$$

where α_k denotes two-digit SIC industry fixed effects. The average ratio captures the level of revenue relative to variable costs, while the coefficient b_{RVC} measures how this ratio varies with firm revenue. Together with the employment-gap moments, these moments help discipline the parameters governing technology investment. Table 6 report the regression results. We see that doubling revenue is associated with approximately 1.76% higher revenue to variable cost ratio.

Table 6: Regression results: Revenue to variable cost ratio

	(1) $\log(R/VC)$
Log revenue	0.0251***
	(0.0056)
Industry fixed effects	Yes
Observations	1,265
R-squared	0.164

Notes: This table report regression results by estimating equation (40). Standard errors in the parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, I use the share of multi-product firms to discipline the product development cost F. This parameter governs the cost of expanding a firm's product portfolio.

To estimate Θ, I minimize the loss function

$$\mathcal{L}(\Theta) \equiv \big(m(\Theta) - \tilde{m}\big)' \mathbf{W} \big(m(\Theta) - \tilde{m}\big),$$

where $m(\Theta)$ is the vector of simulated moments implied by the model, and $\tilde{m}$ is the corresponding vector of moments in the 2020 data. The weighting matrix $\mathbf{W}$ determines the relative importance assigned to each moment in estimation.

6.2.3 Estimation results and model validation

Table 7 reports the estimated parameter values and compares the targeted moments in the model with their empirical counterparts. Panel A lists the targeted moments and their values in the data and the model. The model closely matches moments characterizing the

firm employment-size distribution, employment gaps across products within firms, the fixed-cost share, and how this share varies with firm size. Panel B provides a list of the estimated parameters and their values.

Table 7: Estimated parameters using SMM

Panel A. Targeted Moments	
Moment	Value (Data/Model)
Firm-employment upper-tail slope	1.16 / 1.16
Share multi-product	0.09 / 0.10
Core-to-second product emp gap: Bottom 50% firms	1.02 / 1.02
Core-to-second product emp gap: Top 10% - Bottom 50% firms	0.96 / 0.96
Revenue/variable cost in Compustat (R/VC)	1.55 / 1.48
Regression slope of $\log(R/VC)$ on log sales	0.03 / 0.03

Panel B. Estimated Parameters		
Parameter	Interpretation	Estimate
Firm productivity distribution and fixed cost		
ϑ	Pareto shape parameter	5.866
δ	Speed of productivity decay	0.717
F	Product development cost	0.001
Technology productivity and cost function		
β	IT productivity scale	0.902
ϕ_1	Investment cost scalar	0.011
γ	Investment cost curvature	1.161

Notes: This table reports the results of the Simulated Method of Moments (SMM) estimation. Panel A lists the targeted moments and their values in the data and the model. Panel B reports the estimated parameters and their values.

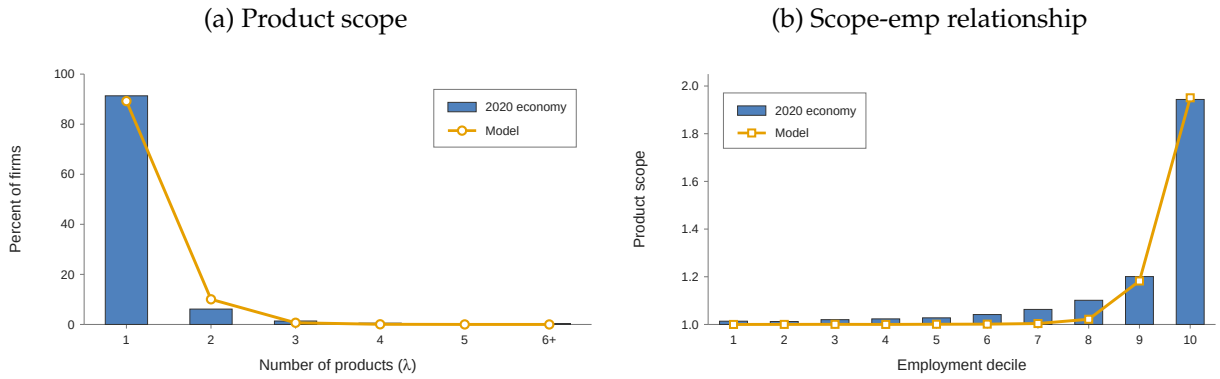
The estimation uses little information about how IT adoption and product scope vary with firm size. I therefore use these untargeted cross-sectional patterns to evaluate the model's quantitative performance.

Figure 2a compares the distribution of firm product scope in the data and the calibrated model. I report the share of firms by the number of products they produce, grouping firms with six or more products into a single category. The blue bars represent the 2020 Dun & Bradstreet data, while the orange line with circle markers shows the corresponding model predictions. The share of single-product firms is targeted in the estimation, while the remaining shares are untargeted. The model closely matches the distribution of product scope, including these untargeted moments.

Figure 2b compares average product scope across firm employment deciles in the model and the 2020 data. The orange line reports the model predictions, while the blue

bars show the corresponding empirical values. Although these cross-sectional patterns are not directly targeted in the estimation, the model closely reproduces the data. For firms in the bottom eight deciles, most produce only one product, a pattern captured by the model. Among firms in the top two deciles, where multi-product firms are more common, the model also closely matches the higher average product scope observed in the data.

Figure 2: Product scope distribution and scope-employment relationship: Model vs. data



Notes: This figure compares the calibrated model with the 2020 Dun & Bradstreet data. Panel (a) plots the distribution of the number of products produced by firms, while Panel (b) plots average product scope by firm employment decile. In both panels, the blue bars represent the data, and the orange line represents the corresponding model predictions.

The average markup is also an untargeted moment. The model predicts a cost-weighted average markup of 1.32 across all producers, which falls within the range of estimates reported in the literature (De Loecker et al. 2020).

7 Quantification Exercise

In this section, I use the quantitative model to evaluate the effects of rise in fixed cost technology on aggregate productivity, and welfare of the whole economy.

7.1 Estimating shocks to the technology efficiency

The main counterfactual examines the effects of changes in the efficiency of fixed-cost technology. To this end, I calibrate the change in the technology-efficiency parameter ψ between 1990 and 2020 so that the model reproduces the observed decline in the fixed-cost share over this period.

Using Compustat data, I calibrate the change in ψ to match the observed percentage change in the aggregate fixed-cost share between 1990 and 2020. The calibration implies that ψ in 1990 is 38.61% lower than its 2020 level ($\psi = 1$).[11]

In Sections 7.1.1 and 7.1.2, I examine how this decline in technology efficiency affects firms' product scope and revenue, as well as economy-wide concentration and welfare. Together, these exercises quantify the extent to which improvements in technology efficiency from 1990 to 2020 account for the observed decline in firm product scope and rise in industry concentration.

7.1.1 Change in firm product scope, and market share

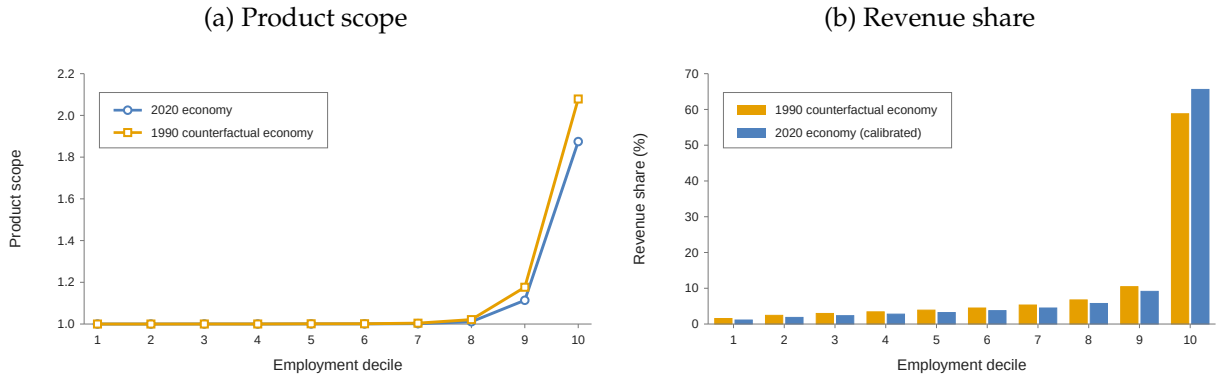
Figure 3a compares average firm product scope across employment deciles in the 1990 counterfactual economy and the calibrated 2020 economy. The orange line reports average product scope under 1990 technology efficiency, while the blue line reports the corresponding values in the 2020 baseline. Two findings stand out. First, average product scope is lower in the 2020 economy throughout the firm-size distribution. Second, the decline is particularly pronounced among the largest firms. For firms in the top employment decile, average product scope is 7.9% lower in the 2020 baseline than in the 1990 counterfactual. This pattern is qualitatively consistent with the 16.7% decline in product scope observed in the data between 1990 and 2020. These results suggest that improvements in technology efficiency are an important driver of the decline in product scope among leading firms.

Figure 3b compares the distribution of revenue across firm employment deciles in the 1990 counterfactual economy and the calibrated 2020 economy. The orange bar reports the revenue share of each employment decile under 1990 technology efficiency, while the blue bar reports the corresponding shares in the 2020 baseline. The revenue share declines for firms in each of the bottom nine deciles, while the top decile gains a larger share of total revenue. Specifically, the revenue share of the top 10% of firms rises from 58.79% in the 1990 counterfactual to 65.60% in the 2020 baseline. This pattern is consistent with the observed increase in the share of industry revenue accounted for by leading firms between 1990 and 2020.

Appendix Figure D.1a and D.1b further plots the average productivity and markup charged by firms in each decile. We see that the productivity and markups are higher in the 2020 baseline, and the gains in productivity and rise in markup are concentrated in the top decile.

[11] The BEA data suggest that the implied quality adjusted unit price goes up by XX during 1990 and 2020.

Figure 3: Product scope and revenue share by firm size



Notes: This figure compares firm product scope and revenue shares across employment deciles in the 1990 counterfactual and the 2020 baseline calibration. In the 1990 counterfactual, I reduce technology efficiency ψ by 38.61% relative to its 2020 level to match the observed change in the aggregate fixed-cost share. Panel (a) plots average product scope by firm employment decile. The blue line represents the 2020 baseline calibration, and the orange line represents the 1990 counterfactual. Panel (b) plots each decile's share of total revenue. The blue bar represents the 2020 baseline calibration, and the orange bar represents the 1990 counterfactual.

7.1.2 Economy-wide concentration, productivity, and welfare

What share of the observed increase in aggregate market concentration can be accounted for by improvements in technology efficiency? Table 8 reports the counterfactual results, comparing the calibrated 2020 economy with an economy in which technology efficiency is reduced to its 1990 level.

First, improvements in technology efficiency generate substantial increases in concentration at both the industry and product levels. Relative to the 1990 counterfactual, the industry HHI in the calibrated 2020 economy is 21.53% higher, while the average product-level HHI increases by 30.29%. The larger increase at the product level suggests that technological progress has a more pronounced effect on concentration within individual products than at the industry level.

Improvements in technology efficiency also affect the number of active firms, both in the aggregate and within individual products. Higher technology efficiency generates stronger selection. Relative to the 1990 counterfactual, the calibrated 2020 economy has 4.91% fewer active firms, reflecting lower participation among less productive firms. This change is consistent with the decline observed in the data, where the number of firms falls by 11.32% between 1990 and 2020, although the model generates a smaller decline. Similarly, the average number of producers per product is 7.73% lower in the calibrated 2020 economy, compared with a decline of 11.55% in the data. Thus, the model generates

declines in the number of active producers both economy-wide and within individual products, consistent with the observed patterns.

Table 8: Concentration, Productivity, and Welfare Change from Technology Change

	Δ Model (%)	Δ Data (%)
A. Industry concentration		
Industry HHI	+21.53 %	–
Observed active firms	-4.91 %	-11.32 %
B. Product-market concentration		
Product market HHI	+30.29 %	–
Suppliers per product: all firms	-3.39 %	–
Suppliers per product: observed firms	-7.73 %	-11.55 %
C. Fixed costs		
Fixed-cost share (%)	+21.88 %	+22.00 %
D. Product scope		
Top-1% mean scope	-15.43 %	-34.89 %
Top-10% mean scope	-7.92 %	-16.15 %
Multi-product share	-24.06 %	-14.68 %
E. Welfare and productivity		
Welfare, $Q = E/P$	+5.72 %	–
Productivity	+8.33 %	–

Notes: This figure compares firm product scope and revenue shares across employment deciles in the 1990 counterfactual and the 2020 baseline calibration. In the 1990 counterfactual, I reduce technology efficiency ψ by 38.61% relative to its 2020 level to match the observed change in the aggregate fixed-cost share. Panel (a) plots average product scope by firm employment decile. The blue line represents the 2020 baseline calibration, and the orange line represents the 1990 counterfactual. Panel (b) plots each decile's share of total revenue. The blue bar represents the 2020 baseline calibration, and the orange bar represents the 1990 counterfactual.

Finally, improvements in technology efficiency generate substantial aggregate productivity and welfare gains. Relative to the 1990 counterfactual, aggregate productivity is 8.33% higher and welfare is 5.72% higher in the calibrated 2020 economy. Thus, although technological progress raises industry concentration, it also generates sizable productivity and welfare gains. This finding is consistent with the view that rising concentration can reflect productivity-enhancing technological change (Autor et al. 2020; Covarrubias et al. 2020).

Importantly, this counterfactual isolates the increase in concentration and markups

driven by improvements in technology efficiency and endogenous technology adoption. Other forces that weaken competition, such as higher barriers to entry, may also raise concentration and markups without generating comparable productivity and welfare gains. These forces are beyond the scope of this paper.

7.1.3 The role of firm adjustments and multi-product firms

The preceding results show that the model generates changes in industry concentration and firm product scope consistent with the data, along with substantial productivity and welfare gains. However, little is known about the sources of these gains. How much comes directly from technological progress, and how much arises from firms' endogenous adjustments in technology investment and product scope? In particular, how important is it to model multiproduct firms rather than single-product firms when quantifying these effects? To address these questions, I conduct additional counterfactual exercises that separate the direct effects of technological progress from those arising through firms' investment and scope adjustments.

To distinguish these margins, I consider three counterfactuals with different restrictions on firms' technology investment and product scope choices. In the first counterfactual, I isolate the direct effect of technological progress by solving for an economy in which technology efficiency increases to its 2020 level, while firms' product scope and technology investment remain fixed at their 1990 levels. Comparing this economy with the 1990 counterfactual allows me to quantify the direct contribution of technological progress to changes in concentration, aggregate productivity, and welfare.

Row 2 of Table 9 reports the change in welfare and productivity in this counterfactual. Technological progress alone raises both aggregate productivity and welfare by 3.75% and 3.95% respectively. These gains account for 45% of the total productivity gain and 69% of the total welfare gain. The remaining 55% and 31%, respectively, arise from firms' endogenous adjustments in technology investment and product scope.

Second, I solve a counterfactual in which firms can reoptimize technology investment while keeping product scope fixed. However, I require productivity growth to be the same across all products within each firm. This restriction approximates a single-product-firm model, in which firms choose their overall technology investment without allowing different productivity gains across products. Row 3 of Table 9 reports the results. Relative to the first counterfactual, endogenous technology investment raises welfare by an additional 0.68% and aggregate productivity by 1.38%. These results show that endogenous investment responses generate additional productivity and welfare gains, even in a

framework that approximates single-product firms.

Third, I allow firms to adjust their product scope while maintaining the restriction of uniform productivity growth across products. This exercise measures the additional effects of endogenous scope adjustment. Row 4 of Table 9 reports the results. Relative to the second counterfactual, allowing firms to adjust their product scope raises welfare by an additional 0.38% and aggregate productivity by 1.21%.[12]

Finally, I return to the full model and allow firms to choose different levels of technology investment across their products. Relative to the third counterfactual, this raises welfare by an additional 0.71% and aggregate productivity by 1.99%. Firms allocate more investment toward their most productive products and reduce investment in peripheral ones. These adjustments allow more productive producers to expand production, generating additional productivity and welfare gains.

This result is related to the gains from production reallocation studied by Baqaee and Farhi (2020). In my model, however, firms not only reallocate production toward more productive suppliers but also endogenously improve their production technologies. By selectively investing in their stronger products, firms change both the distribution of productivity and the allocation of production across products. These adjustments further increase aggregate productivity and welfare.

Comparing the full model with the second counterfactual allows me to quantify the additional gains from endogenous product scope and heterogeneous technology investment across products. These two margins are unique to the multiproduct setting and help assess the importance of modeling multiproduct firms rather than single-product firms. Overall, firms' endogenous adjustments generate additional welfare gains of 1.77 percentage points and productivity gains of 4.58 percentage points. Of these gains, allowing firms to adjust their product scope and invest selectively across products accounts for 1.09 percentage points of welfare and 3.20 percentage points of productivity. These represent 20.9% and 38.3% of the total welfare and productivity gains, respectively. Thus, abstracting from the multiproduct nature of firms would understate the aggregate productivity and welfare effects of improvements in fixed-cost technology efficiency.

[12]The overall level of technology investment also changes relative to the second counterfactual.

Table 9: Decomposing Welfare and Productivity Changes

	Welfare	Productivity
Baseline: 1990 equilibrium	+0.00% (—)	+0.00% (—)
CF 1. Technology improvement; adoption frozen	+3.95% (+3.95%)	+3.75% (+3.75%)
CF 2. Firm-wide adoption; scope fixed	+4.63% (+0.68%)	+5.13% (+1.38%)
CF 3. Scope + firm-wide adoption	+5.01% (+0.38%)	+6.34% (+1.21%)
Full adjustment: 2020 equilibrium	+5.72% (+0.71%)	+8.33% (+1.99%)

7.2 Policy counterfactual

7.2.1 Subsidizing product creation

I first consider a policy aimed at encouraging the creation of new product varieties. Specifically, I consider a subsidy covering half of firms' product-development costs. The motivation is that the private return to developing a new variety may differ from its social return. Firms do not internalize the benefits to consumers from greater variety and stronger competition. The former can lead to too few varieties (Dhingra 2013; Mankiw and Whinston 1986).[13] These forces make the welfare effects of subsidizing product creation ambiguous.

Specifically, in the counterfactual, the subsidy applies uniformly to expenditure on the entire chosen portfolio. It lowers firms' private development costs without changing the labor required for development. The government finances its payments through a lump-sum tax on the representative household. I hold the calibrated parameters, technology efficiency, and mass of potential firms fixed and re-solve the equilibrium, allowing firms to adjust participation, product scope, technology investment, and prices. The exercise therefore evaluates whether additional variety and changes in competition justify the resources devoted to product development and any crowding out of technology investment in existing products.

Column (1) in Table 10 reports the policy counterfactual results. In the calibrated model, subsidizing product creation reduces welfare. Two forces explain this result. First, expanding product scope crowds out technology investment in firms' core products, lowering overall production productivity. Second, developing additional products requires more labor, leaving fewer resources available for production. Together, these effects reduce aggregate welfare.

[13] On the other hand, firms do not internalize business-stealing effects on competing producers, potentially resulting in excessive supply of varieties.

Table 10: Policy Counterfactual

	Development subsidy (1)	Markup-targeting transfer (2)
Welfare	−1.05%	+2.17%
Productivity	−0.21%	+3.55%

7.2.2 Policy analysis: Subsidy on high markup products

I next consider a product-level tax-and-subsidy schedule designed to correct the misallocation arising from markup dispersion, following Edmond et al. (2023). Conditional on technologies and product-market participation, high-markup suppliers produce too little relative to lower-markup suppliers. This creates differences in the marginal social returns to production labor across suppliers. Reallocating labor can therefore raise aggregate output without increasing total production labor. The relevant distortion is markup dispersion, rather than concentration itself. If high-markup suppliers are also more productive, correcting this distortion may increase their market shares and raise concentration.

The policy induces a common price-to-marginal-cost ratio equal to the baseline cost-weighted aggregate markup. It eliminates dispersion in pricing wedges while preserving the baseline aggregate markup, rather than uniformly subsidizing sales. I hold the calibrated parameters, technology efficiency, and mass of potential firms fixed. I then solve for a new equilibrium, allowing firms to adjust their product scope and technology investment.

Column (2) of Table 10 reports the counterfactual results. In the calibrated model, eliminating markup dispersion raises aggregate productivity by 2.17% and welfare by 3.55%. These results indicate that markup dispersion generates substantial allocative losses in the calibrated economy.

8 Conclusion

(Pending writeup)

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A Data Appendix

A.1 Dun & Bradstreet Data

A.2 Nielsen Retail Scanner Data

Following Hottman et al. (2016), I link individual barcodes to firms using GS1 firm identifiers. The Nielsen product classification does not map directly into SIC industries. I therefore define industries using Nielsen's seven broad product categories and product markets using Nielsen product groups. A firm's product scope is measured as the number of distinct product groups in which it sells at least one barcode.[14]

A.3 Example products produced documented in data

Table A.1: List of products produced by Intel in electronic equipment manufacturing

Product code	Short definition	1990	1995	2000	2005	2010	2015	2020
36740201	Computer logic modules	✓	✓	✓	✓	✓	✓	✓
36740204	Solid-state memory	✓	✓	✓	✓	✓	✓	✓
36740207	Microprocessors	✓	✓	✓	✓	✓	✓	✓
36740211	Semiconductor circuit networks	✓	✓	✓	✓	–	–	✓
36770000	Electronic coils/transformers	✓	–	–	–	–	–	–
36790102	Electronic commutators	✓	–	–	–	–	–	–
36720000	Printed circuit boards	✓	–	–	–	–	–	–
36740206	Integrated microcircuits	–	–	✓	✓	✓	✓	✓
36740200	Integrated circuits/networks	–	–	✓	✓	✓	✓	–
36610000	Telephone/telegraph equipment	–	–	✓	–	–	–	–
36619908	Fiber-optic communications equipment	–	–	–	✓	✓	✓	–
Product scope		7	4	7	7	6	6	5

[14] Within a product market, I interpret the collection of barcodes supplied by a firm as that firm's variety. As a robustness check, I define product markets using the more disaggregated Nielsen product modules and obtain similar results.

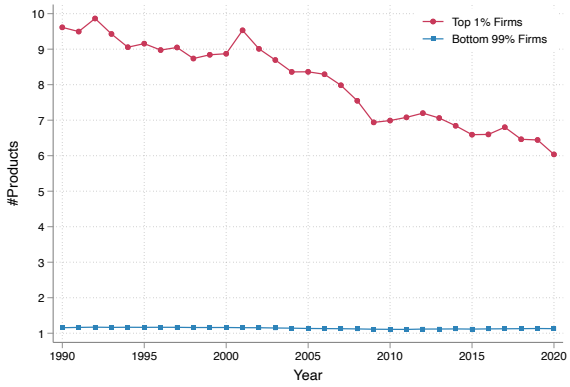
B Additional Stylized Facts

B.1 Product scope of top 20 firms

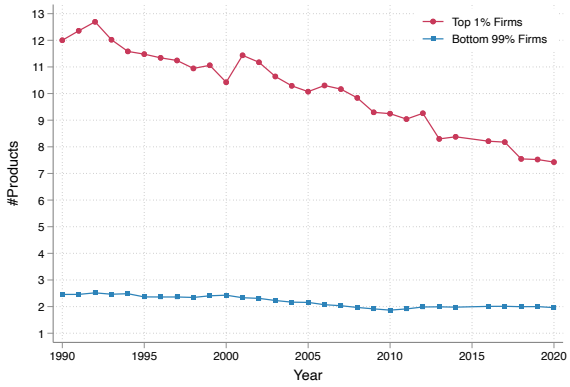
In the baseline, I define industry leaders as the top 1% of firms by employment within each industry. However, the number of firms in each industry changes over time, which can affect the composition and size of the top 1% group. In this section, I examine whether the decline in product scope is robust to an alternative definition of industry leaders. Following Autor et al. (2020), I define leading firms as the top 20 firms by employment in each industry and year. Figure B.1 shows that the decline in product scope is similar to that reported in the main text. Thus, the results are robust to defining industry leaders using a fixed number of firms rather than a fixed percentile of the employment distribution.

Figure B.1: Product Scope of Top 20 Firms

(a) Top 20 Firms: Unweighted



(b) Top 20 Firms: Emp weighted



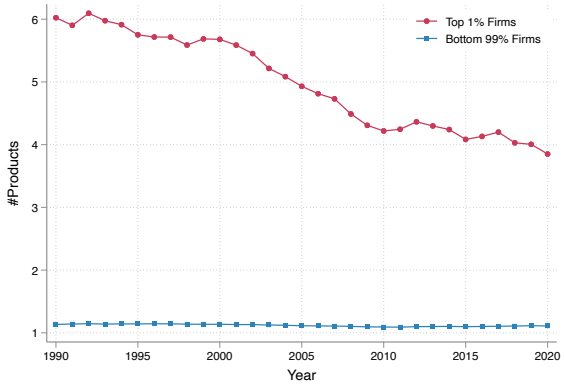
Notes: This figure plots the average number of products produced by the top 20 firms from 1990 to 2020. For each year, I first compute average product scope among the top 1% of firms within each two-digit SIC industry. I then aggregate these industry-level averages across manufacturing industries, using each industry's share of total manufacturing employment in 1990 as a fixed weight in both panels. Panel (a) plots the resulting aggregate series when firms are equally weighted within each industry, while Panel (b) plots the corresponding series when firms are weighted by their employment. We see that there is a substantial decline in the product scope of top 20 firms.

B.2 Product scope of top 1%, Variable Industry Weights

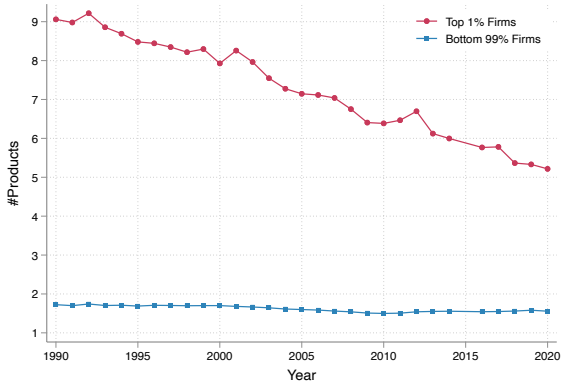
In the baseline, I aggregate industry-level measures of product scope among the top 1% of firms using each industry's share of total manufacturing employment in 1990 as a fixed weight. However, industry employment shares change over time. In this section, I examine whether the results are sensitive to this choice by allowing industry weights to vary each year. Specifically, I use each industry's share of total manufacturing employment in that year. Figure B.2 shows that both the trend and magnitude of the decline in product scope are similar to those reported in the main text. Thus, the decline in product scope among leading firms is not driven by the use of fixed 1990 industry weights.

Figure B.2: Product Scope of Top 1% Firms, Variable Industry Weights

(a) Top 1% Firms: Unweighted



(b) Top 1% Firms: Emp weighted



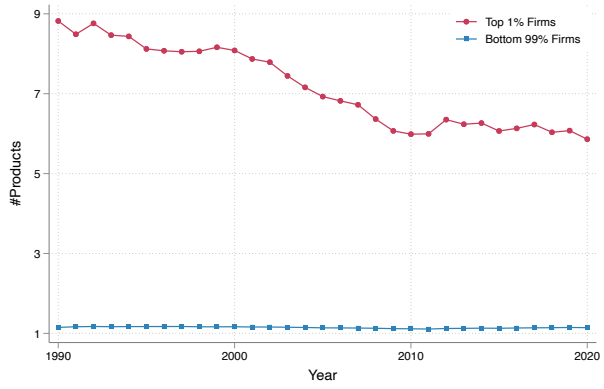
Notes: This figure plots the average number of products produced by the top 1% of firms from 1990 to 2020. For each year, I first compute average product scope among the top 1% of firms within each two-digit SIC industry. I then aggregate these industry-level averages across manufacturing industries, using each industry's share of total manufacturing employment in every year as weight in both panels. Panel (a) plots the resulting aggregate series when firms are equally weighted within each industry, while Panel (b) plots the corresponding series when firms are weighted by their employment. We see that there is a substantial decline in the product scope of top 1% firms.

B.3 Product scope of top 1% Firms, alternative firm definition

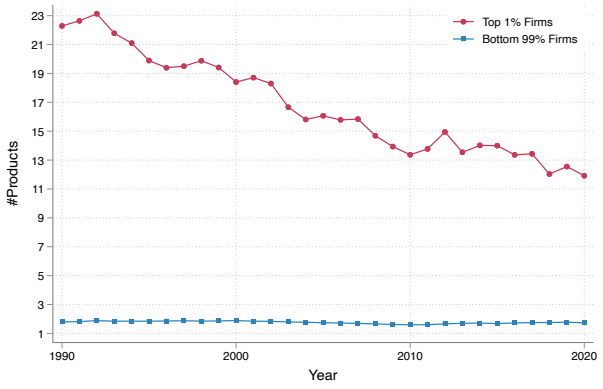
In this section, I examine whether the decline in product scope is robust to an alternative firm definition. In the baseline, I define a firm as the set of establishments sharing the same headquarters DUNS number within a two-digit SIC industry. Here, I instead group all establishments with the same headquarters DUNS number into a single firm and count its products across all manufacturing industries. Figure B.3 plots the average product scope of the top 1% of firms under this broader definition. Panel (a) shows that the unweighted average falls from nearly 9 products in 1990 to fewer than 6 products in 2020. Panel (b) shows a similar decline in employment-weighted average product scope. These results confirm that the decline in product scope among leading firms persists even when their products are counted across all manufacturing industries.

Figure B.3: Product Scope of Top 1% Firms

(a) Top 1% Firms: Unweighted



(b) Top 1% Firms: Emp weighted



Notes: This figure plots the average number of products produced by the top 1% of firms from 1990 to 2020. Here, I instead group all establishments with the same headquarters DUNS number into a single firm and count its products across all manufacturing industries. For each year, I compute average product scope among the top 1% of firms within manufacturing sector. Panel (a) plots the resulting aggregate series when firms are equally weighted within each industry, while Panel (b) plots the corresponding series when firms are weighted by their employment. We see that there is a substantial decline in the product scope of top 1% firms.

B.4 Product scope results using Nielsen Scanner Dataset

We further check the number of products produced by a firm using the Nielsen Scanner Dataset. Nielsen Scanner data provide alternative definition of a product market.

B.5 Other stylized facts

B.5.1 Declining product-market overlap among leading firms

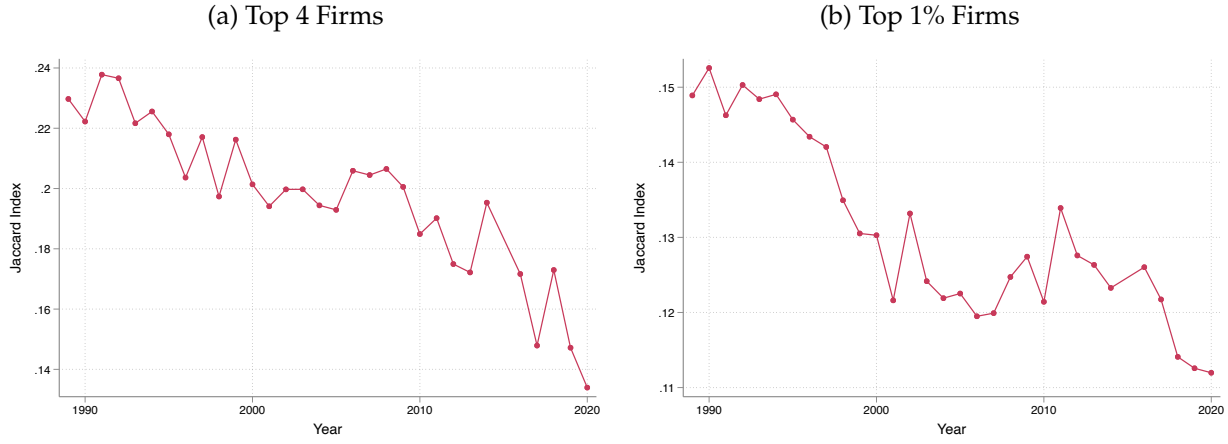
I next examine whether leading firms increasingly operate in the same product markets. For each pair of firms i and i' within the same industry, let S_i and $S_{i'}$ denote the sets of products produced by the two firms. I measure product-market overlap using the Jaccard index,

$$J(S_i, S_{i'}) = \frac{|S_i \cap S_{i'}|}{|S_i \cup S_{i'}|}.$$

This index equals zero when two firms produce no common products and equals one when they produce exactly the same set of products. For both the top four firms and the top 1 percent of firms within each industry, I compute the average bilateral Jaccard index across all firm pairs.

Figure B.4 plots the average Jaccard index from 1990 to 2020. Panel (a) reports the series for the top four firms within each industry, while panel (b) reports the corresponding series for firms in the top 1 percent of the within-industry employment distribution. Product-market overlap among leading firms declines steadily over time. This pattern indicates that leading firms are not increasingly converging toward the same broad product portfolios. Instead, their product portfolios are becoming less similar, consistent with a process of specialization in which top firms narrow their product scope and concentrate on more distinct sets of products.

Figure B.4: Average bilateral Jaccard index



Notes:

B.6 Summary statistics, raw product data

Here I report the summary staitistics for the product market without dropping products with less than 20 producers.

Table B.1: Summary statistics on firm product scope

Sample	# Unique products		Firms per product				Employment per product			
	N	Per industry	Mean	Median	P90	P99	Mean	Median	P90	P99
1988–2018, Full sample										
All products	14,508	216.0	99.9	12	166	1,632	6,136.3	584.4	8,835.2	99,685.9
Manufacturing	8,473	423.7	23.1	6	44	274	2,102.2	364.5	4,255.0	28,223.5
Services	1,969	149.7	235.7	48	500	3,596	14,735.6	1,742.0	25,470.6	204,943.7
1988 Sample										
All products	14,496	213.2	83.8	11	137	1,280	4,917.5	534.7	7,710.5	78,234.4
Manufacturing	8,630	431.5	24.7	6	47	289	2,433.4	408.1	4,932.0	32,255.1
Services	1,906	136.1	147.1	28	296	2,136	9,519.6	889.3	15,250.5	142,726.5
2018 Sample										
All products	14,350	214.2	107.6	11	172	1,883	7,252.8	570.4	10,171.5	119,364.0
Manufacturing	8,276	413.8	21.0	6	39	270	1,991.3	307.0	3,745.3	29,451.5
Services	1,979	152.2	280.1	51	630	3,912	18,950.4	2,179.0	35,479.5	302,295.1

B.7 Firm scope increase with IT intensity

I first examine the cross-sectional relationship between firms' product scope and their use of information technology. Specifically, I estimate,

$$y_{ikt} = \sum_{r=1}^{10} \beta_r \, \mathbb{1}\{\mathrm{IT}_{ikt} = r\} + \alpha_k + \gamma_t + \varepsilon_{ikt}, \tag{41}$$

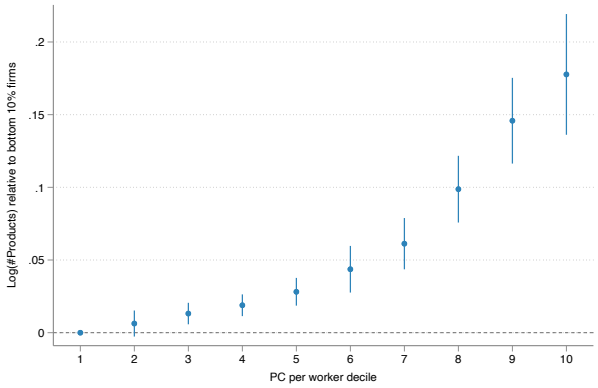
where y_{ikt} denotes the log number of products produced by the firm i in industry k and year t. The variable IT_{ikt} denotes firm i's decile of information-technology intensity within industry k and year t, with the first decile omitted as the reference group. The coefficients β_r therefore compare firms in higher IT-intensity deciles to firms in the lowest decile, within the same industry and year. The industry fixed effects α_k absorb permanent differences in product scope across industries, while the year fixed effects γ_t absorb aggregate changes in product scope over time.

I measure IT intensity using two alternative proxies. The first is the log number of computers per worker, following Bloom et al. (2012). The second is log IT spending per worker, following Lashkari et al. (2024). The coefficients of interest are the β_r's, which trace out how product scope varies with firms' IT intensity within narrowly defined industry-year cells.

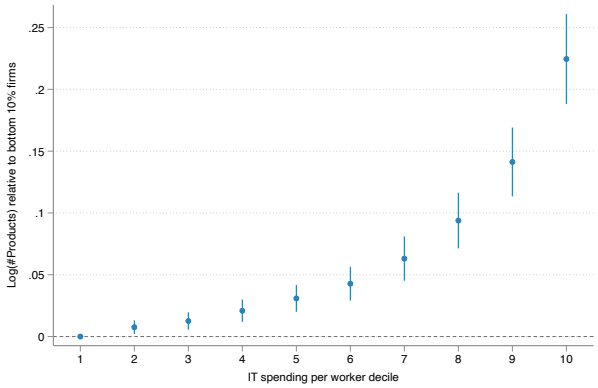
Figure B.5 plots the estimated coefficients β_r from equation (41). In Panel (a), I estimate the specification using PCs per worker to construct firms' IT-intensity deciles. I find that product scope increases monotonically across the IT-intensity distribution: firms with higher IT intensity operate in a larger number of product markets. The magnitude is economically meaningful. Firms in the top decile of PCs per worker operate 18 log points more products than firms in the bottom decile, conditional on industry and year fixed effects. Panel (b) repeats the same exercise using IT spending per worker to construct firms' IT-intensity deciles. I again find a monotonic relationship between IT intensity and product scope. Firms in the top decile of IT spending per worker operate 22 log points more products than firms in the bottom decile. Thus, across both measures, firms with greater IT intensity have broader product scope within industries.

Figure B.5: Product scope increases with IT intensity

(a) IT measure: PC per worker



(b) IT measure: IT spending per worker



Notes:

C Theory Appendix

C.1 Equilibrium definition

Definition 1 (Symmetric Bayesian Equilibrium) A symmetric Bayesian equilibrium consists of, for each industry k, of a first-stage scope policy function $\lambda^k(A)$, a distribution G^k of first-stage firm characteristics (A_i^k, λ_i^k), a family $\Gamma^k = \left\{\Gamma^k(\lambda)\right\}_{\lambda \in \Lambda^k}$ of scope-contingent competitive environments, Stage-2 price, investment, and quantity $\{p_{ij}^k, x_{ij}^k, q_{ij}^k\}_{i \in \mathcal{I}^k}$ for all firms and their products, and an industry price index P^k. Together with aggregate income Y and price index P. These objects satisfy the following conditions.

1. *Stage-2 optimality.* For every realization generated by G^k and the product assignment process, firms' price and investment choices $\{p_{ij}^k, x_{ij}^k\}$ form a Nash equilibrium (15).
2. *Stage-1 optimality.* The stage-1 scope policy function solves the firm problem (17).
3. *Consistency of firm characteristics and competitive environments.* The distribution G^k is induced by the exogenous productivity distribution of A and the scope policy functions $\lambda^k(A)$. For each $\lambda \in \Lambda^k$, $\Gamma^k(\lambda)$ is the distribution of payoff-relevant environments generated when the focal firm chooses product scope λ. The distributions of competitive environments used in Stage-1 problems coincide with the distribution generated by firms' equilibrium choices and the product market assignment process.
4. *Price-index and goods-market consistency.* Product-market and industry price indices satisfy (6) and (7). Quantities satisfy the demand equations (4) and (8). Production follows $q_{ij}^k = z_{ij}^k l_{ij}^k$.
5. *Labor-market clearing.* Investment and product development costs are paid in units of labor. Let

$$X_i^k \equiv \sum_{j \in \mathcal{J}_i^k} x_{ij}^k$$

denote firm i's total technology investment in Stage 2. With the wage normalized to one, aggregate labor supply $\bar{L}$ satisfies

$$\bar{L} = \sum_{k=1}^{K} N^k \int \left\{ F\lambda + \mathbb{E}_{\omega \sim \Gamma^k(\lambda)} \left[\sum_{j \in \mathcal{J}_i^k} c_{ij}^k q_{ij}^k + C\left(X_i^k\right) \right] \right\} dG^k(A, \lambda). \tag{42}$$

6. *Aggregate income.* Aggregate income is

$$Y = \bar{L} + \Pi, \tag{43}$$

where aggregate profits are

$$\Pi = \sum_{k=1}^{K} N^k \int \left\{ \mathbb{E}_{\omega \sim \Gamma^k(\lambda)} \left[\sum_{j \in \mathcal{J}_i^k} \left(p_{ij}^k - c_{ij}^k \right) q_{ij}^k - C\left(X_i^k \right) \right] - F\lambda \right\} dG^k(A, \lambda). \tag{44}$$

The equilibrium definition therefore closes three fixed points: firms' first-stage choices determine the distribution of realized competitive environments; those environments determine Stage-2 prices and investment decisions; and the resulting allocations determine the aggregate price indices and income taken as given by individual firms.

C.2 Proof of Lemma 1

C.3 Proof of Proposition 1

C.4 Model Mechanism under Oligopolistic Competition.

Optimal product-level investment. Consider next the firm's product-specific fixed-cost technology investment choice. Let $X_i^k = \sum_{j \in \mathcal{J}_i^j} x_{ij}^k$ and $\nu_i^k = C'\left(X_i^k\right) = \phi_1 \gamma (X_i^k)^{\gamma-1}$ denote total investment and its common marginal cost.

Since $\frac{\partial c_{ij}^k}{\partial x_{ij}^k} = -\psi c_{ij}^k g'\left(\psi x_{ij}^k\right)$, the envelope theorem applied to the firm's optimal pricing decision gives

$$\psi q_{ij}^k c_{ij}^k g'\left(u_{ij}^k\right) = \psi l_{ij}^k g'(u_{ij}^k) \leq \nu_i^k, \qquad u_{ij}^k = \psi x_{ij}^k, \quad x_{ij}^k \geq 0. \tag{45}$$

with equality whenever $x_{ij}^k > 0$.

Two forces govern this investment condition. First, the benefit from investment depends on the product's production base. The term $q_{ij}^k c_{ij}^k$ is the product's total variable cost. Because labor is the only variable input and the wage is normalized to one, this term also equals production employment, l_{ij}^k. The left-hand side of equation (45) therefore measures the labor savings generated by additional product-level investment.

Holding the marginal investment cost fixed, investment is more valuable for a product with a larger production base. A producer with higher fundamental productivity, $A_i a_{ij}$, operates at a larger scale and therefore has a stronger incentive to invest. Higher investment further lowers its marginal cost and expands its production base, reinforcing the incentive to invest. Fixed-cost technology investment therefore amplifies initial differences in fundamental productivity and favors producers that are already more productive.

Importantly, the production base within each market is not determined by the firm's own productivity alone. It also depends on competitors' productivity, investment, and pricing decisions, which determine the firm's competitive position in that market. Competition therefore directly shapes the benefit from investment on the left-hand side of equation (45).

The second force operates on the cost side. Optimal investment in each product depends on the common marginal investment cost, ν_i^k, which rises with the firm's total investment when $\gamma > 1$. Greater investment in one product raises the marginal cost of investment throughout the portfolio, inducing the firm to reduce investment elsewhere. In this sense, products within the firm compete for investment resources through the common marginal cost. The firm allocates investment by equating marginal returns across products receiving positive investment. Consequently, investment in a given product depends not only on conditions in its own market, but also on the firm's investment opportunities elsewhere in its portfolio.

Optimal product scope. Consider next the firm's product scope choice. The firm chooses scope before its assignments and competitive environment are realized. To isolate the internal investment competition trade-of, I first consider the value of an additional product conditional on a realized sequence of opportunities, and then return to the expected Stage-1 comparison.

Let $J_{i,n}$ denote the set of product markets associated with firm i's first n product opportunities, with $|\mathcal{J}_{i,n}| = n$. The next opportunity serves market $j_{i,n+1} \notin_{i,n}$, so that $\mathcal{J}_{i,n+1} = \mathcal{J}_{i,n} \cup \{j_{i,n+1}\}$.

Let $\pi_{ij}(x)$ denote operating profit in market j after optimal pricing, but before investment and product-development costs. The firm's optimized profit with n products, net of both investment costs and product creation costs, is

$$W_i^{(n)} = \max_{\{x_{ij}\geq 0\}_{j\in\mathcal{J}_{i,n}}} \left\{ \sum_{j\in\mathcal{J}_{i,n}} \pi_{ij}(x_{ij}^{(n)}) - C\left(\sum_{j\in\mathcal{J}_{i,n}} x_{ij}^{(n)} \right) - nF \right\}. \tag{46}$$

where $x_{ij}^{(n)}$ is the optimal investment under scope n, and define

$$X_i^{(m)} \equiv \sum_{j\in\mathcal{J}_{i,n}} x_{ij}^{(n)}, \qquad \nu_i^{(n)} \equiv C'(X_i^{(n)}), \qquad \pi_{ij}^{(n)} \equiv \pi_{ij}\left(x_{ij}^{(n)}\right).$$

Define $D_{i,n} \equiv W_i^{(n+1)} - W_i^{(n)}$ as the net incremental profit from increasing scope from n to $n+1$. If n is optimal, adding the next product is unprofitable and $D_{i,n} < 0$. The gains from adding product $n+1$, which sells in market $j_{i,n+1}$ is

$$D_{i,n} = \pi_{i,j_{i,n+1}}^{(n+1)} - F - \left[C(X_i^{(n+1)}) - C(X_i^{(n)})\right] - \sum_{h\in\mathcal{J}_{i,n}} \left[\pi_{ih}^{(n)} - \pi_{ih}^{(n+1)}\right]. \tag{47}$$

Adding a product generates additional operating profit but also incurs a development cost and raises the firm's total investment expenditure. When total investment rises, the common marginal cost of investment increases, crowding out investment in existing products. The final term captures the resulting loss of operating profits from these products. Thus, the value of adding a product depends not only on its own profitability but also on the adjustments it induces elsewhere in the firm's portfolio.

To characterize this linkage, I use a small-product approximation. For a sufficiently large portfolio, adding one product can therefore be approximated as a marginal expansion of scope. Investment in the additional product is small relative to investment in the existing portfolio. The following lemma characterizes the resulting investment crowding

out and incremental product value.

Lemma 2 (Marginal product value and internal investment competition)**.** *Define*

$$\epsilon_{D,ih} \equiv -\frac{\partial \log x_{ih}(\nu)}{\partial \log \nu} > 0, \qquad \bar{\epsilon}_{D,i} \equiv \sum_{\substack{h \in \mathcal{J}_{i,n} \\ x_{ih}^{(n)} > 0}} \frac{x_{ih}^{(n)}}{X_i^{(n)}} \epsilon_{D,ih}, \qquad \iota_i \equiv \frac{(\gamma-1)\bar{\epsilon}_{D,i}}{1+(\gamma-1)\bar{\epsilon}_{D,i}}.$$

Then, to first order around the initial n-product allocation[15]*,*

1. *Adding the product crowds out investment in each existing product h by*

$$-\Delta \log x_{ih} \approx \iota_i \frac{\epsilon_{D,ih}}{\bar{\epsilon}_{D,i}} \frac{x_{i,j_{i,n+1}}^{(n+1)}}{X_{i,n}}, \tag{48}$$

   where $\Delta \log x_{ih} \equiv \log x_{ih}^{(n+1)} - \log x_{ih}^{(n)}$*. Consequently,*

$$-\sum_{h \in \mathcal{J}_{i,n}} \Delta x_{ih} \approx \iota_i x_{i,j_{i,n+1}}^{(n+1)}, \qquad X_i^{(n+1)} - X_i^{(n)} \approx (1-\iota_i) x_{i,j_{i,n+1}}^{(n+1)}.$$

2. *The net value of producing the additional product is*

$$D_{i,n} \approx \pi_{i,j_{i,n+1}}^{(n+1)} - F - \underbrace{(1-\iota_i)\nu_i^{(n)} x_{i,j_{i,n+1}}^{(n+1)}}_{\text{additional investment expenditure}} - \underbrace{\iota_i \nu_i^{(n)} x_{i,j_{i,n+1}}^{(n+1)}}_{\text{forgone operating profit}} \tag{49}$$

Lemma 2 makes the breadth–depth trade-off explicit. Equation (48) characterizes how adding a new product crowds out investment in existing products. For each unit invested in the new product, investment in existing products falls by approximately ι_i units in total. Products whose investment is more sensitive to the marginal investment cost experience larger proportional reductions. When $\gamma > 1$, expanding scope therefore comes partly at the expense of technological upgrading in existing products. This raises their unit costs and weakens their market positions. Conversely, narrowing scope relaxes this pressure and supports deeper investment in the retained products.

Equation (49) separates the development cost, the increase in total investment expenditure, and operating profits forgone on existing products. The first term, F, captures the

[15]For this local calculation, competitors' prices and aggregate demand conditions are held at their initial equilibrium values while the focal firm reoptimizes its prices and investment. The expected Stage-1 comparison below uses equilibrium continuation values under each scope choice and includes the associated competitive responses.

cost of creating one additional product. The second term $\nu_i(1-\iota_i)x_{i,j_{i,n+1}}^{(n+1)}$, captures the increase in the firm's total investment cost. The third term, $\nu_i\iota_i x_{i,j_{i,n+1}}^{(n+1)}$, captures the operating profits forgone on existing products due to investment crowding out.

Returning to Stage 1, let $V(A_i,\omega)$ denote equilibrium Stage-2 profit net of investment costs but before product-development costs. Evaluated at the equilibrium associated with a realization ω under scope n, the net payoff defined above is $W_i^{(n)}(\omega) = V(A_i,\omega) - nF$. The expected gain from choosing $n+1$ rather than n products is

$$\begin{aligned}\overline{D}_{i,n} &\equiv \mathbb{E}_{\omega\sim\Gamma(n+1)}\left[W_i^{(n+1)}(\omega)\right] - \mathbb{E}_{\omega\sim\Gamma(n)}\left[W_i^{(n)}(\omega)\right] \\ &= \mathbb{E}_{\omega\sim\Gamma(n+1)}[V(A_i,\omega)] - \mathbb{E}_{\omega\sim\Gamma(n)}[V(A_i,\omega)] - F.\end{aligned} \tag{50}$$

The firm chooses scope to maximize expected net profit. Thus, $\bar{D}_{i,n} < 0$ means thaat adding the next product is unprofitable relative to retaining n products.

C.5 Response to technology shock

I next consider a small, common improvement in technology efficiency around an arbitrary positive value of ψ. This allows me to examine how better scalable technology changes firms' incentives to expand within existing products or across more products. The analysis helps explain why leading firms can capture larger shares of industry activity while producing fewer products.

Proposition 2 (Better technology, product deepening, and scope)**.** *Consider a small improvement in technology efficiency ψ, and holding industry expenditure fixed. Suppose g is increasing and strictly concave, $\kappa(u) = [g'(u)]^2/[-g''(u)]$ is strictly increasing, and $ug'(u)$ is non-decreasing. Define,*

$$G_{ij} \equiv \frac{d\log z_{ij}}{d\log\psi}, \qquad \overline{G}_j \equiv -\frac{d\log p_j}{d\log\psi}, \qquad \overline{G}_P \equiv -\frac{d\log P}{d\log\psi}.$$

(i) (Scale-biased upgrading and market-share selection.) For two suppliers in the same market,

$$x_{ij} > x_{\ell j} \iff \frac{l_{ij}}{\nu_i} > \frac{l_{\ell j}}{\nu_\ell}. \tag{51}$$

Their market shares respond according to

$$\frac{d\log s_{ij}}{d\log\psi} = \varsigma(s_{ij})(G_{ij} - \overline{G}_j), \qquad \varsigma(s) = \frac{\eta-1}{1+(\eta-1)\dfrac{(\eta-\sigma)s}{\varepsilon(s)[\varepsilon(s)-1]}} > 0. \tag{52}$$

(ii) (Marginal product value and internal amplification.) Let $j = j_{i,n+1}$ be the additional product and let $\widetilde{D}_{i,n} = \max_{x\geq 0}\{\pi_{ij}(x) - F - \nu_i x\}$ denote the marginal-product approximation underlying Lemma 2. The marginal value of adidtional product is approximately optimized contribution satisfies

$$\frac{1}{\pi_{ij}}\frac{d\widetilde{D}_{i,n}}{d\log\psi} = \underbrace{(\varepsilon_{ij}-1)u_{ij}g'(u_{ij}) - \left[(\eta-\sigma)\sum_{\ell\in\mathcal{M}_j\setminus\{i\}} s_{\ell j}\left(G_{\ell j} - \frac{d\log\mu_{\ell j}}{d\log\psi}\right) + (\sigma-1)\overline{G}_P\right]}_{\text{product-market selection}} - \underbrace{\frac{\nu_i x_{ij}}{\pi_{ij}}\frac{d\log\nu_i}{d\log\psi}}_{\text{internal investment competition}}. \tag{53}$$

where

$$\frac{d\log\nu_i}{d\log\psi} = (\gamma-1)\sum_{h\in\mathcal{J}_{i,n}}\omega^x_{ih}\frac{d\log x_{ih}}{d\log\psi}, \qquad \omega^x_{ih} = \frac{x_{ih}}{X_i}. \tag{54}$$

Proposition 2 identifies two forces governing how better technology changes the value of the marginal product: product-market competition and internal competition for investment. The first captures changes in a product's operating profit as its own technology and its competitors' technologies improve. The second captures how investment elsewhere in the firm's portfolio affects its marginal investment cost and the value of adding another product.

Product-market competition operates through scale-dependent returns to technology investment. As equation (45) shows, an investment that reduces unit labor requirements generates greater savings when applied to a larger production base. Under the increasing-$\kappa(u)$ condition, better technology reinforces this scale advantage. Holding the parent's marginal investment cost fixed, larger producers invest more and obtain greater productivity gains. These uneven gains strengthen the competitive positions of larger producers while making weaker producers less competitive. A product can become more productive yet lose market share if its productivity gain falls below the market benchmark, $G_{ij} < \overline{G}_j$. As a result, better technology can therefore reduce the value of retaining a marginal product even before accounting for internal investment competition.

This selection mechanism is related to the product-mix adjustments studied by Bernard et al. (2011) and Mayer et al. (2014). In those settings, changes in the trading environment induce firms to adjust their product portfolios and withdraw marginal products. Here, competitive pressure arises endogenously from technological upgrading. Larger suppliers have stronger incentives to invest, making products with smaller relative productivity

gains less attractive.

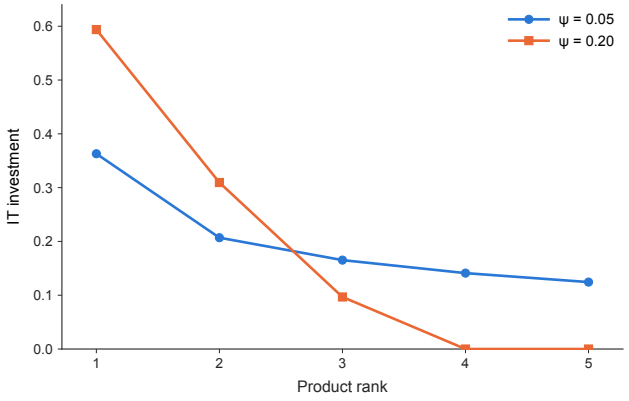
In addition to the selection effects, internal competition for investment can further reduce the value of marginal products when better technology increases investment in the firm's existing products. Because investment costs are convex, this expansion raises the common marginal investment cost, $\nu_i = C'(X_i)$, faced by all products. Equation (54) shows that the increase depends on product-level investment responses weighted by their shares in total firm investment. Greater investment in the firm's largest product lines can therefore raise the cost of investing in other products.

The higher marginal investment cost reduces the value of adding another product. As equation (53) shows, this effect can reinforce losses from product-market competition or outweigh gains from technological upgrading. Importantly, it suggests that a product may remain a strong competitor in its own market yet become unattractive to its parent firm because of more valuable investment opportunities elsewhere in the portfolio.

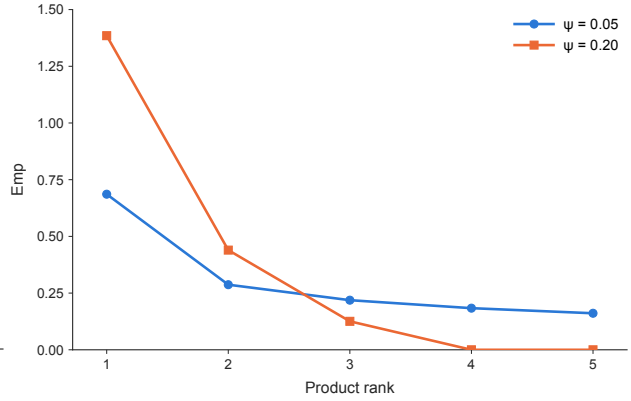
These two forces together reduce scope when they make the expected incremental contribution of another opportunity insufficient to cover its development cost. A decline in $\widetilde{D}_{i,n}$ describes the marginal-product mechanism; the actual Stage-1 decision compares the full expected equilibrium contributions $\overline{D}_{i,n}$. At an adjacent indifference boundary, a decline in $\overline{D}_{i,n}$ favors the narrower portfolio. Away from that boundary, the product can become less valuable without an immediate scope adjustment. Figure C.1 provides a graphical representation of this result. We assume the productivity from fixed-cost technology is $g(u) = \bar{g}\log(1+u^{\beta})$, which satisfies the proposition's shape restrictions at positive investment levels.

Figure C.1: Model mechanism

(a) Fixed-cost investment



(b) Employment size

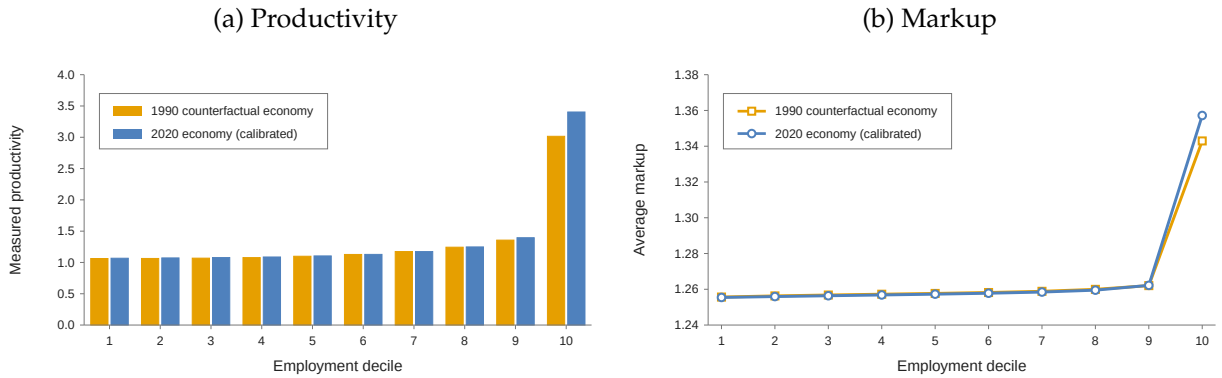


Notes:

C.6 Proposition 3.

D Additional Quantification Results

Figure D.1: Productivity and markup by firm size



Notes: This figure compares firm productivity and markup across employment deciles in the 1990 counterfactual and the 2020 baseline calibration. In the 1990 counterfactual, I reduce technology efficiency ψ by 38.61% relative to its 2020 level to match the observed change in the aggregate fixed-cost share. Panel (a) plots average productivity by firm employment decile, the blue bar represents the 2020 baseline calibration, and the orange bar represents the 1990 counterfactual. Panel (b) plots each decile's average markup. In both figures, the blue line represents the 2020 baseline calibration, and the orange line represents the 1990 counterfactual.